

# 1 Questions

**Nick's Question 1.1** In the Wave equation paper you showed that for the Greens function, we have that

$$\mathcal{F}G(t, \cdot)(\xi) = \frac{\sin(t|\xi|)}{|\xi|}$$

and so we see that

$$\mathcal{F}G(t, \cdot)(c\xi) = \frac{1}{c} \mathcal{F}(ct, \cdot)(\xi)$$

However in the Fractional SPDE paper the greens function is more complicated and we have that

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} H_{1,2}^{1,1} \left( 2^{-1} \nu t^\beta |\xi|^\alpha \middle| \begin{matrix} (0, 1) \\ (0, 1) \end{matrix} \right. (1 - \beta - \gamma, \beta) \Big) \\ &= t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_{1/2}^1(s) \left( 2^{-1} \nu t^\beta |\xi|^\alpha \right)^{-s} ds \end{aligned}$$

And so we see that

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_{1/2}^1(s) \left( 2^{-1} \nu t^\beta |c\xi|^\alpha \right)^{-s} ds$$

However I am unsure how to handle the constant,  $c$ , on the inside so we can get a nice scaling property like we do in the Wave Equation paper.

Also, instead of using the Fox-H function we can just say

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

and so

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))}$$

but I still don't see how to handle the  $c$ .

Here is how one should proceed:

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta c^\alpha |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} (tc^{\frac{\alpha}{\beta}})^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

$$= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} \mathcal{F}Y_{\alpha,\beta,\gamma,d} \left( c^{\frac{\alpha}{\beta}} t, \cdot \right) (c\xi)$$

Please check if this scaling reduces to heat and wave equations correctly.

**Nick's Question 1.2** In the wave equation paper with Xia Chen you said that that fundamental solution to the SPDE

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) - \frac{1}{2} \Delta u(t, x) = \sqrt{\Theta} u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 \\ u_t(0, \cdot) = 0 \end{cases} \quad (1.1) \quad \text{wspde}$$

and the mild solution is defined to be

$$u(t, x) = 1 + \sqrt{\Theta} \int_0^t \left( \int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds \quad (1.2) \quad \text{weqs}$$

but for general initial conditions for the wave equation we have the solution to be

$$\begin{aligned} u(t, x) = \frac{\partial}{\partial t} \int_{\mathbb{R}^d} G(t-s, x-y) u(0, y) dy + \int_{\mathbb{R}^d} G(t-s, x-y) u_t(0, y) dy \\ + \sqrt{\Theta} \int_0^t \left( \int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds \end{aligned} \quad (1.3) \quad \text{gweqs}$$

But when the initial conditions are such as in 1.1 then the first two integral in 1.3 are 1 and 0 respectively and so we arrive at 1.2.

In the fractional SPDE

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma [\rho(u(t, x)) \dot{W}(x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu & \beta \in (0, 1] \\ u(0, \cdot) = \mu_0, \quad u_t(0, \cdot) = \mu_1 & \beta \in (1, 2] \end{cases} \quad (1.4) \quad \text{fspde}$$

and we define

$$J_0(t, x) = \begin{cases} (Z(t, \cdot) * \mu)(x) & \beta \in (0, 1] \\ (Z^*(t, \cdot) * \mu_0)(x) + (Z(t, \cdot) * \mu_1)(x) & \beta \in (1, 2] \end{cases}$$

and

$$I(t, x) = \iint_{[0, t] \times \mathbb{R}} Y(t-s, x-y) \rho(u(s, y)) W(\delta y) ds$$

and the general solution for 1.4 is

$$u(t, x) = J_0(t, x) + I(t, x) \quad (1.5) \quad \text{feqs}$$

My question is that if we consider the following fractional spde

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma [\rho(u(t, x)) \dot{W}(x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2] \end{cases} \quad (1.6) \quad \text{fspde1}$$

then can we calculate  $J_0(t, x)$ ? More specifically is  $J_0(t, x) = 1$  in this case? The wave equation paper worked out nicely because in that paper  $J_0(t, x) = 1$  and so the solution was able to be written as 1.2. The 1 allows us to easily do the Piccard iteration to get equation (2.3) from the Wave equation paper with Xia Chen.

So if our solution can not be written as

$$u(t, x) = 1 + I(t, x)$$

I am not sure at the moment how we can come up with an expression for the  $f'_n$ s in the Weiner Expansion like you did in equation (2.3) in the paper with Xia Chen.

Hi, Nick, here is the answer:

Notice that, by Theorem 4.1 of my paper,

$$\begin{aligned} (Z_{\alpha, \beta, \gamma, d}(t, \cdot) * 1)(x) &= \mathcal{F}Z_{\alpha, \beta, \gamma, d}(t, \cdot)(0) \\ &= t^{\lceil \beta \rceil - 1} E_{\beta, \lceil \beta \rceil}(0) \\ &= \frac{t^{\lceil \beta \rceil - 1}}{\Gamma(\lceil \beta \rceil)}. \end{aligned}$$

Hence, when  $\beta \in (0, 1]$ , we see that  $(Z_{\alpha, \beta, \gamma, d}(t, \cdot) * 1)(x) = 1$ . When  $\beta \in (1, 2]$ ,

$$\begin{aligned} (Z_{\alpha, \beta, \gamma, d}^*(t, \cdot) * 1)(x) &= \mathcal{F}Z_{\alpha, \beta, \gamma, d}^*(t, \cdot)(0) \\ &= E_{\beta}(0) \\ &= \frac{1}{\Gamma(1)} = 1. \end{aligned}$$

Hence, under the initial conditions given by (1.6),  $J_0(t, x) \equiv 1$ .

**Nick's Question 1.3 Guess for the constant (1.20) from the Wave Equation Paper** In the wave equation paper, the constant (1.20) was

$$C = \begin{cases} \frac{4 - \Delta}{3 - \Delta} & \text{under Assumption A} \\ \frac{4 - d}{3 - d} & \text{under Assumption B or C} \end{cases} \quad (1.7) \quad \boxed{\text{const}}$$

I believe in the fractional paper the constant will be

$$\begin{cases} \frac{2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha}{2(\beta + \gamma - 1) + 1 - \beta\Delta/\alpha} & \text{under Assumption A} \\ \frac{2(\beta + \gamma - 1) + 2 - \beta d/\alpha}{2(\beta + \gamma - 1) + 1 - \beta d/\alpha} & \text{under Assumption B or C} \end{cases}$$

I think this because in the wave equation paper where we use  $(4 - \Delta)$  this gets replaced by  $(2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha)$  and thus we can guess that

$$\begin{cases} 4 - \Delta & \rightarrow & 2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha \\ 3 - \Delta & \rightarrow & 2(\beta + \gamma - 1) + 1 - \beta\Delta/\alpha \end{cases}$$

Note that when  $\alpha = \beta = 2$  and when  $\gamma = 0$  then the guess reduces to the wave equation.

**Nick's Question 1.4 Lemma 2.3 for the fractional equation**

1. In the outline you sent me, you use

$$\mathcal{F}_a = \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\}$$

Where  $H^a(\mathbb{R}^d)$  is the fractional Sobelov Space where  $H^2(\mathbb{R}^d) = W^{1,2}(\mathbb{R}^d)$ . However, I saw on line that  $H^a(\mathbb{R}^d) = W^{a,2}(\mathbb{R}^d)$  and so  $H^2(\mathbb{R}^d) = W^{2,2}(\mathbb{R}^d)$ . Is there a mistake here?

2. Where is  $\mathcal{M}$  defined with the sup over  $\mathcal{F}_d$  and not  $\mathcal{F}_a$ ? Is this a typo?
3. When defining  $\mathcal{E}_a(g, g)$  why does the  $(2\pi)^{-d}$  appear here but not in the wave paper?
4. How do we generalize  $\rho$  from the wave paper to the fractional equation? For the wave equation we have that

$$\rho = (\mathcal{M}(\gamma, 2))^{(4-\alpha)/2}$$

but why does it generalize to this in the fractional sense

$$\rho = (\mathcal{M}_a(\gamma, 2))^{(2a-\alpha)/a} = 2^{-a/\alpha} (\mathcal{M}_a(\gamma, 1))^{(2a-\alpha)/a}$$

which is what you put in the outline you sent me.

**Nick's Question 1.5 : Typos from Xia Chen Paper (Large deviations for Riesz potentials of additive processes)**

From what I can tell there seem to be several typos that I can't justify from this paper that seem serious. The following are some definitions and a Theorem from Xia Chen's Paper.

**Definitions:**

$$\rho = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f(s+t)f(t)}{\sqrt{1+|s+t|^\beta} \sqrt{1+|t|^\beta}} dt \right] \varphi_{d-\alpha}(s) ds \quad (1.8)$$

$$\mathcal{E}_\beta(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |s|^\beta |\hat{f}(s)|^2 ds \quad (1.9)$$

$$\mathcal{F}_\beta = \left\{ f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1, \mathcal{E}_\beta(f, f) < \infty \right\} \quad (1.10)$$

$$\Lambda_\alpha = \sup_{f \in \mathcal{F}_\beta} \left\{ \left( \int_{\mathbb{R}^d} \frac{\prod_{j=1}^p f^2(x_j)}{|x_1 + \dots + x_p|^\alpha} \prod_{j=1}^p dx_j \right)^{1/p} - \mathcal{E}_\beta(f, f) \right\} \quad (1.11)$$

**Theorem 1.5**

$$\rho = (2\pi)^{-d} (\Lambda_\alpha)^{p-(\alpha/\beta)}$$

This theorem is proved in section 7 of the paper and starts with two lemmas.

**Lemma 7.1:** Lemma 7.1 claims that  $\Lambda_\alpha$  is finite, however I believe that there is a typo in the proof. In the very first line of the proof, it references equation (A.2) and (A.2) is

proved by referencing (A.8). I believe that equation (A.8) is wrong. (A.8) says that

$$\int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p f_j(x_j)}{|x_1 + \dots + x_p|^{d-\alpha}} \prod_{j=1}^p dx_j = \int_{\mathbb{R}^{d2}} \frac{f_1(x)(f_2 * \dots * f_p)(y)}{|x - y|^{d-\alpha}} dx dy$$

but if we let  $p = 2$  then (A.8) says that

$$\int_{\mathbb{R}^{2d}} \frac{f_1(x_1)f_2(x_2)}{|x_1 + x_2|^{d-\alpha}} dx_1 dx_2 = \int_{\mathbb{R}^{d2}} \frac{f_1(x)f_2(y)}{|x - y|^{d-\alpha}} dx dy$$

which is not true. For this issue, we might need to assume that all  $f_i$  are symmetric functions.

### Lemma 7.2

From first glance I think Lemma 7.2 is okay.

**Proof of Theorem 1.5:** First some definitions for this theorem.

**Definitions:**

1.

$$L(f) = \left( \int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j \right)^{1/p}$$

2.

$$\tilde{L}(f) = \mathcal{E}_\beta(f, f)$$

3.

$$J = \sup_{\|f\|_2^2 + \tilde{f} = 1} L(f) = \sup_{f \in \mathcal{H}} \frac{L(f)}{\|f\|_2^2 + \tilde{L}(f)}$$

There are several places in the proof that I think are typos. Also I do not understand for sure why we have the equality in the supremums in definition (3) above.

1. First is equation (7.21) which says by Parseval's Relation we get

$$\sup_{\langle g, Qg \rangle = 1} \int_{\mathbb{R}^d} [(Qg) * (\widetilde{Qg})]^p(-s) \varphi(s) ds = \sup_{\langle g, Qg \rangle = 1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}([(Qg) * (\widetilde{Qg})]^p)(x) \mathcal{F}\varphi(x) dx \quad (1.12)$$

typo1

where  $\tilde{p}(x) = p(-x)$  and  $g(x) = f(x)\sqrt{1 + |x|^\beta}$  where  $f$  is the same function from the definition of  $\rho$ . Notice the left hand side of 1.12 has a  $(-s)$  as the input of the function. Then when taking the Fourier transform of  $[(Qg) * (\widetilde{Qg})]^p(-s)$  we should have the following

$$\mathcal{F}[(Qg) * (\widetilde{Qg})]^p(-x)$$

which follows from the property that if we let  $\tilde{p}(x) = p(-x)$  then  $\mathcal{F}(\tilde{p})(\xi) = \mathcal{F}(p)(-\xi)$ .

Here is what is going on here: First of all, for  $\tilde{p}(x) = p(-x)$ ,

$$\mathcal{F}\tilde{p}(\xi) = \int_{\mathbb{R}^d} e^{-ix\xi} p(-x) dx = \int_{\mathbb{R}^d} e^{ix\xi} p(x) dx = \overline{\mathcal{F}f(\xi)}.$$

Hence,

$$\int_{\mathbb{R}^d} \tilde{f}(x)g(x)dx = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}g(\xi) \overline{\mathcal{F}\tilde{f}(\xi)} d\xi = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}g(\xi) \mathcal{F}f(\xi) d\xi.$$

**Nicks Solution:** The centered equation (7.20) in the Xia Chen papers says

$$\int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} (Qg)(s+t)(Qg)(t) dt \right]^p \varphi(s) ds = \int_{\mathbb{R}^d} \left[ (Qg) * (\widetilde{Qg}) \right]^p (-s) \varphi(s) ds$$

and this equality is true, but we can continue in a different way. Using a substitution  $u = s + t$  we see that

$$\begin{aligned} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} (Qg)(s+t)(Qg)(t) dt \right]^p \varphi(s) ds &= \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} (Qg)(u)(Qg)(u-s) du \right]^p \varphi(s) ds \\ &= \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} (Qg)(u)(\widetilde{Qg})(s-u) du \right]^p \varphi(s) ds \\ &= \int_{\mathbb{R}^d} \left[ (Qg) * (\widetilde{Qg}) \right]^p (s) \varphi(s) ds \end{aligned}$$

and now we can apply Parseval's to see that

$$\int_{\mathbb{R}^d} \left[ (Qg) * (\widetilde{Qg}) \right]^p (s) \varphi(s) ds = (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F} \left[ (Qg) * (\widetilde{Qg}) \right]^p (x) \mathcal{F}\varphi(x) dx$$

and this gives us equation (7.21). What this actually showed was

$$s \mapsto \int_{\mathbb{R}^d} (Qg)(s+t)(Qg)(t) dt$$

is an even function no matter what  $(Qg)$  by just applying the substitution  $u = s + t$ .

2. I'm going to continue, assuming that the previous typo was actually not a typo. The next typo is the unlabeled centered equation right above (7.22) which says

$$\begin{aligned} \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F} \left[ (Qg) * (\widetilde{Qg}) \right]^p (x) \mathcal{F}\varphi(x) dx & \quad (1.13) \quad \boxed{\text{typo2}} \\ &= \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d(p+1)} \int_{\mathbb{R}^d} [|\widehat{Qg}|^2]^{*p} (x) \mathcal{F}\varphi(x) dx \end{aligned}$$

It is stated in the paper that  $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$  and  $\mathcal{F}(fg) = (2\pi)^{-d}\mathcal{F}(f) * \mathcal{F}(g)$  and  $f^{*p}$  is the  $p$ -fold convolution. So using these rules we should get that

$$\begin{aligned} \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F} \left( \left[ (Qg) * (\widetilde{Qg}) \right]^p \right) (x) \mathcal{F}\varphi(x) dx \\ &= \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d} \int_{\mathbb{R}^d} (2\pi)^{-(p-1)d} \mathcal{F} \left( \left[ (Qg) * (\widetilde{Qg}) \right] \right)^{*p} (x) \mathcal{F}\varphi(x) dx \\ &= \sup_{\langle g, Qg \rangle=1} (2\pi)^{-dp} \int_{\mathbb{R}^d} \left[ \widehat{Qg} \widehat{\widetilde{Qg}} \right]^{*p} (x) \mathcal{F}\varphi(x) dx \end{aligned}$$

Xia's arguments are correct:

$$\mathcal{F} \left( f * \widetilde{f} \right) (\xi) = \mathcal{F}(f)(\xi) \mathcal{F}(\widetilde{f})(\xi) = \mathcal{F}(f)(\xi) \overline{\mathcal{F}(f)(\xi)} = |\mathcal{F}(f)(\xi)|^2.$$

It seems as if they are assuming that  $\widehat{\widehat{Qg}} = \widehat{Qg}$  which is only true if  $Qg$  is an even function.

**Nicks Solution:** Since  $Qg$  is a real valued function then we have the property that

$$\widehat{\widehat{Qg}} = \overline{\widehat{Qg}}$$

and therefore we have that

$$\int_{\mathbb{R}^d} \left[ \widehat{Qg} \widehat{\widehat{Qg}} \right]^{*p}(x) \mathcal{F}\varphi(x) dx = \int_{\mathbb{R}^d} \left[ |\widehat{Qg}|^2 \right]^{*p}(x) \mathcal{F}\varphi(x) dx$$

which verifies equation (7.22).

3. **Details for (7.22)** After fixing the (7.22) typo we will have the following. In giving the details for (7.22) just for clarity. (7.22) says that if we define  $h = (2\pi)^{-d} \widehat{Qg}$  then

$$\begin{aligned} & \sup_{\langle g, Qg \rangle = 1} (2\pi)^{-dp} \int_{\mathbb{R}^d} [|\widehat{Qg}|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx \\ &= \sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1} (2\pi)^{dp} \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx \end{aligned}$$

Since  $(2\pi)^d h = \widehat{Qg}$  then this implies that  $\hat{h} = \widetilde{Qg} = Q\tilde{g}$  since  $Q$  is even and because when considering  $\hat{f}(x) = \int \exp(ixs) f(s) ds$  as the definition of the Fourier transform then we have the property that  $\hat{\hat{f}} = (2\pi)^d \tilde{f}$ . Also we have that

$$\langle g, Qg \rangle = \langle \tilde{g}, \widetilde{Qg} \rangle = \langle \hat{h} Q^{-1}, \hat{h} \rangle = (2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h)$$

since for this definition of the Fourier transform we have  $\langle f, g \rangle = (2\pi)^{-d} \langle \hat{f}, \hat{g} \rangle$ . Putting this together gives us (7.22).

Here are the arguments: we may assume  $g$  to be even functions, indeed the supremum is achieved for even functions  $g$ . Now

$$\begin{aligned} \langle g, Qg \rangle &= \langle \tilde{g}, Qg \rangle \\ &= \langle \bar{\hat{h}}, Q^{-1} \hat{h} \rangle \\ &= \langle \bar{\hat{h}}, (1 + \psi) \hat{h} \rangle \\ &= \|\hat{f}\|_2^2 + \int_{\mathbb{R}^d} (1 + \psi(\xi)) |\hat{h}(\xi)|^2 d\xi \\ &= \|\hat{f}\|_2^2 + \int_{\mathbb{R}^d} (1 + \psi(\xi)) |\hat{h}(\xi)|^2 d\xi \\ &= (2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h). \end{aligned}$$

4. **Equation (7.23)** Details for the following after fixing the (7.22) typo.

$$\sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1} (2\pi)^{dp} \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx$$

$$= \sup_{\|h\|_2^2 + \tilde{L}(h)=1} \int_{R^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j$$

In particular how the sup limits changed.

*Possible Justification:* The integrals can be shown to be equal through a change of variables.

$$\begin{aligned} \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx &= \int_{\mathbb{R}^d} (|h|^2 * \dots * |h|^2)(x) \frac{1}{|x|^\alpha} dx \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^{d(p-1)}} |h|^2(x - y_1) |h|^2(y_1 - y_2) \dots |h|^2(y_{p-2} - y_{p-1}) |h|^2(y_{p-1}) \prod_{j=1}^{p-1} dy_j \frac{1}{|x|^\alpha} dx \end{aligned}$$

then applying the substitution

$$\begin{pmatrix} x - y_1 = u_1 \\ y_1 - y_2 = u_2 \\ \dots \\ y_{p-2} - y_{p-1} = u_{p-1} \\ y_{p-1} = u_p \end{pmatrix}$$

will transform the integral as

$$\int_{\mathbb{R}^d} (|h|^2 * \dots * |h|^2)(x) \frac{1}{|x|^\alpha} dx = \int_{R^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j$$

As for the limits of the supremum, note that

$$(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1 \implies \left\| (2\pi)^{d/2} h \right\|_2^2 + \tilde{L}((2\pi)^{d/2} h) = 1$$

Therefore using the above definitions of  $J$ ,  $L(f)$  and  $\tilde{L}(f)$  we see that

$$\begin{aligned} \sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h)=1} (2\pi)^{dp} \int_{R^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j &= \sup_{\left\| (2\pi)^{d/2} h \right\|_2^2 + \tilde{L}((2\pi)^{d/2} h)=1} (2\pi)^{dp} L(h)^p \\ &= (2\pi)^{dp} \left[ \sup_{\left\| (2\pi)^{d/2} h \right\|_2^2 + \tilde{L}((2\pi)^{d/2} h)=1} L(h) \right]^p \\ &= (2\pi)^{dp} \left[ \sup_{h \in \mathcal{H}} \frac{L(h)}{\left\| (2\pi)^{d/2} h \right\|_2^2 + \tilde{L}((2\pi)^{d/2} h)} \right]^p \\ &= (2\pi)^{dp} \left[ (2\pi)^{-d} \sup_{h \in \mathcal{H}} \frac{L(h)}{\|h\|_2^2 + \tilde{L}(h)} \right]^p \\ &= J^p \end{aligned}$$

which is exactly equation (7.23) in the Xia Chen paper.

Because of equation (7.17) we should have that

$$\rho = \Lambda(1)^{p-\sigma/\beta} \quad (1.14) \quad \text{rhofix}$$

### How to apply the above to correlations functions other than the Reisz Kernal:

Suppose we have a spectral measure,  $\mu$ , with spectral density  $\varphi$  with related tempered measure  $\Gamma$  with density  $\gamma$ . In other words, for all Schwartz functions  $\psi$

$$\begin{aligned} \int_{\mathbb{R}^d} dx \gamma(x) \psi(x) &= \int_{\mathbb{R}^d} \Gamma(dx) \psi(x) \\ &= \int_{\mathbb{R}^d} \mu(dx) \mathcal{F}\psi(x) \\ &= \int_{\mathbb{R}^d} dx \varphi(x) \mathcal{F}\psi(x) \end{aligned}$$

also suppose that we have the following scaling properties

$$\gamma(cx) = c^{-\alpha} \gamma(x) \quad \text{and} \quad \varphi(cx) = c^{-(d-\alpha)}$$

We can follow the proof of Theorem 1.5 from the Xia Chen paper with minor modifications to show that

$$\rho = (2\pi)^{-d} \Lambda(1)^{2-\alpha/\beta}$$

where

$$\rho = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f(s+t)f(t)}{\sqrt{1+|s+t|^\beta} \sqrt{1+|t|^\beta}} dt \right] \varphi(s) ds$$

and

$$\Lambda(\theta) = \theta^{\beta/(\beta-\alpha/2)} \Lambda(1)$$

and

$$\begin{aligned} \Lambda(1) &= \sup_{\|f\|_2=1} \left\{ \left( \int_{\mathbb{R}^{2d}} |f(x_1)|^2 |f(x_2)|^2 \gamma(x_1+x_2) dx_1 dx_2 \right)^{1/2} - \mathcal{E}_\beta(f, f) \right\} \\ &= \sup_{\|f\|_2=1} \left\{ \left( \int_{\mathbb{R}^d} [|f|^2 * |f|^2](x) \gamma(x) dx \right)^{1/2} - \mathcal{E}_\beta(f, f) \right\} \\ &= \sup_{\|f\|_2=1} \left\{ \langle |f|^2 * |f|^2, \gamma \rangle^{1/2} - \mathcal{E}_\beta(f, f) \right\} \end{aligned}$$

If this is correct then this means that equation (1.7) in the wave paper is slightly is slightly off.

### Nick's Question 1.6 Addressing the (A.2) and (A.8) Typo from Xia Chen's Paper

(A.2) says the following, for  $0 < \sigma < d$  and some constant  $C$ ,

$$\left| \int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p f_j(x_j)}{|x_1 + \dots + x_p|^{d-\sigma}} \prod_{j=1}^p dx_j \right| \leq C^p \prod_{j=1}^p \|f_j\|_{pd/((p-1)d+\sigma)}$$

and the proof of this inequality relies on (A.8) which says

$$\int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p f_j(x_j)}{|x_1 + \dots + x_p|^{d-\alpha}} \prod_{j=1}^p dx_j = \int_{\mathbb{R}^{d2}} \frac{f_1(x)(f_2 * \dots * f_p)(y)}{|x - y|^{d-\alpha}} dx dy$$

and (A.8) is only true for even functions. In the Xia Chen paper, (A.2) is used to prove (A.3) as well however these equations are only used in two spots.

For (A.2), it is used to show that  $\Gamma_\sigma$  is finite. It seems like this is unnecessary because in section 7 if Chen's paper, it is proven that  $\rho = (2\pi)^{-d}(\Gamma(1))^{p-\sigma/\beta}$  and  $\rho$  had already been proven to be finite in the beginning of the paper and so then  $\Gamma$  must be finite. So using (A.2) to independently prove that  $\Gamma$  is finite seems unnecessary at the moment but I am unsure.

(A.3), and also (A.2), is used in the proof of Lemma 2.2 which says that

$$\mathbb{E} [\zeta([0, t_1], \dots, [0, t_p])^m] = \int_{\mathbb{R}^{dm}} \left[ \sum_{\sigma \in \Sigma_m} \prod_{k=1}^m \frac{1}{\lambda_{\sigma(1)+\dots+\lambda_{\sigma(k)}}} \right]^p \prod_{k=1}^m \varphi(\lambda_k) d\lambda_k.$$

It seems that once we have the expression on the right hand side that the proofs for the upper and lower bound in Theorem 2.3 work fine. So maybe in our paper this wont be an issue since we do not deal with the left hand side of the above centered equality.

**Possible fix for the (A.2) Inequality :** In the Wave paper, we only care about  $\rho$  from Xia Chen's paper with  $p = 2$ . We can rewrite (A.2) as

$$\int_{\mathbb{R}^{2d}} \frac{f(x)g(y)}{|x + y|^{d-\sigma}} dx dy = \int_{\mathbb{R}^{2d}} \frac{f(u)g(-v)}{|u - v|^{d-\sigma}} dx dy = \int_{\mathbb{R}^{2d}} \frac{f(u)\tilde{g}(v)}{|u - v|^{d-\sigma}} dx dy$$

Then we can apply (1.26) from Xia Chen's paper with  $r = 2d/(\sigma + d)$  and  $s = 2d/(\sigma + d)$  to obtain

$$\int_{\mathbb{R}^{2d}} \frac{f(u)\tilde{g}(v)}{|u - v|^{d-\sigma}} dx dy \leq C \|f\|_{2d/(\sigma+d)} \|\tilde{g}\|_{2d/(\sigma+d)}$$

but

$$\|\tilde{g}\|_{2d/(\sigma+d)} = \|g\|_{2d/(\sigma+d)}$$

and so

$$\int_{\mathbb{R}^{2d}} \frac{f(x)g(y)}{|x + y|^{d-\sigma}} dx dy \leq C \|f\|_{2d/(\sigma+d)} \|\tilde{g}\|_{2d/(\sigma+d)} = C \|f\|_{2d/(\sigma+d)} \|g\|_{2d/(\sigma+d)}$$

and this is the inequality we need but only for  $p = 2$ .

### Nick's Question 1.7 1.26 From The Xia Chen Paper

I am unsure why we can say

$$\int_{\mathbb{R}^{2d}} \frac{F(\lambda)G(\gamma)}{|\lambda - \gamma|^{d-\sigma}} d\lambda d\gamma \leq C \|F\|_r \|G\|_s$$

where  $s^{-1} + r^{-1} = 1 + \sigma/d$ . It says it follows by page 275 of *Donoghue - Distributions and Fourier transforms* but in that text what is stated is that

$$\int_{\mathbb{R}^{2d}} u(\lambda)v(\gamma)T(\lambda - \gamma)d\lambda d\gamma \leq \|T\|_r \|v\|_{r'} \|u\|_1$$

where I am unsure what  $r$  and  $r'$  satisfy but  $T$  is required to be in  $L^r(\mathbb{R}^d)$  and in the case

in Xia Chen's paper,  $T(x) = \varphi(x) = \frac{1}{|x|^{d-\sigma}}$  is not in  $L^r \mathbb{R}^d$ . I think in the Donoghue text we have  $1/r + 1/r' + 1 = 2$  which would follow from Young's Convolution Inequality.

The equation (1.26) can be written as

$$\int_{\mathbb{R}^d} I_\sigma(F)(\gamma) G(\gamma) d\gamma \leq C \|F\|_r \|G\|_s$$

where

$$I_\sigma(F)(\gamma) = \int_{\mathbb{R}^d} \frac{F(x)}{|\gamma - x|^{d-\sigma}} dx$$

and I tried using the theorem (Hardy Littlewood Fractional Integration Theorem) which says that  $I_\sigma$  maps from  $L^p$  to  $L^q$  if and only if  $1/p - 1/q = \sigma/d$  with  $p > 1$  but as of now I am still working on it.

**Partial Solution:** Let  $q$  and  $s$  be such that  $1/q + 1/s = 1$  then Holders Inequality gives us that

$$\begin{aligned} \int_{\mathbb{R}^{2d}} \frac{F(\lambda) G(\gamma)}{|\lambda - \gamma|^{d-\sigma}} d\lambda d\gamma &= \int_{\mathbb{R}^d} I_\sigma F(\lambda) G(\gamma) d\lambda \\ &\leq \|I_\sigma F\|_q \|G\|_s \end{aligned}$$

Now recalling the Hardy Littlewood Fractional Integral Theorem,  $I_\sigma$  is a bounded linear operator from  $L_r$  to  $L_q$  if and only if  $1/q = 1/r - \sigma/d$ . Let  $r$  be such. Then we have

$$\|I_\sigma F\|_q \leq C \|F\|_r$$

and putting this together we get that

$$\int_{\mathbb{R}^{2d}} \frac{F(\lambda) G(\gamma)}{|\lambda - \gamma|^{d-\sigma}} d\lambda d\gamma \leq C \|F\|_r \|G\|_s$$

whenever

$$\begin{aligned} 1 &= \frac{1}{q} + \frac{1}{s} \\ &= \left( \frac{1}{r} - \frac{\sigma}{d} \right) + \frac{1}{s} \end{aligned}$$

hence

$$\frac{1}{r} + \frac{1}{s} = 1 + \frac{\sigma}{d}$$

which is what is given in the Xia Chen paper.

Your solutions above look very good.

**Nick's Question 1.8 (1.26) for the fractional noise.**

If we want to apply the results to

$$\varphi(x) = \prod_{j=1}^d \frac{1}{|x|^{1-\sigma_j}}$$

then we need to be able to prove equation (1.26) for this noise which is

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \frac{F(\gamma)G(\lambda)}{\prod_{j=1}^d |\lambda_j - \gamma_j|^{1-\sigma_j}} d\gamma d\lambda \leq C \|F\|_r \|G\|_s$$

In order to figure that how we need to find out properties about the following operator,

$$L_\sigma f(x) = \int_{\mathbb{R}^d} \frac{f(y)}{\prod_{j=1}^d |x_j - y_j|^{1-\sigma_j}} dy.$$

Specifically can we say that  $L_\sigma$  is a bounded linear operator from  $L_p$  to  $L_q$  for some related pair  $(p, q)$ ?

The issue with copying the proof for the Hardy-Littlewood fractional integral theorem in this situation is that polar coordinates do not seem to be so easily applied in this situation.

#### Weak Young's Inequality:

The inequality (1.26) from the Xia Chen paper seems to only be important to show that  $\rho$  is finite. We may be able to use Young's Weak inequality to do this, which is

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x)g(x-y)g(y)dx dy \right| \leq K_{p,q,r,d} \|f\|_p \|g\|_{q,w} \|h\|_r$$

Where  $p, q, r > 1$  and  $1/p + 1/q + 1/r = 2$ . I am following page 107 section 4.3 of Analysis by Lieb and Loss. The weak  $L^q$  space is defined to be the set of functions such that

$$\sup_{\alpha>0} \alpha |\{x : |f(x)| > \alpha\}|^{1/q} < \infty.$$

This quantity is not a norm on the  $L_w^q$  space though. But there is a norm on the space that it is equivalent to the above expression (I am unsure what equivalent means in this situation since this statement is not proven in the Lieb and Loss text)

$$\|f\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |f(x)| dx$$

where  $1/q + 1/q' = 1$  and  $A$  is a measurable set of finite measure.

It is also claimed in the Lieb and Loss text that

$$\|f\|_{d/\lambda,w} = \frac{d}{d-\lambda} \left[ |\mathbb{S}^{d-1}|/d \right]^{\lambda/d} \quad \text{when } f(x) = |x|^{-\lambda}$$

If we use this but with  $\lambda = d - \alpha$  and  $q = d/\lambda$  then we get that  $1/p + 1/r = 1 + \alpha/d$  and also

$$\int_{\mathbb{R}^{2d}} \frac{f(x)g(x)}{|x-y|^{d-\lambda}} dx dy \leq K \frac{d}{\alpha} \left[ |\mathbb{S}^{d-1}|/d \right]^{(d-\alpha)/d} \|f\|_p \|g\|_r.$$

This is another way to prove that  $\rho$  from the Xia Chen paper is finite and this way avoids using the Sobelov-Hardy-Littlewood Theorem that I used above. Things need some work here because we need  $\|f\|_p$  and  $\|g\|_r$  to be finite. In the Xia Chen paper you can see that these norms just reduced to  $L^2$  norms because they have those functions  $F$  and  $G$  and when they reduce to  $f$  and  $g$  then the norms become  $L^2$  norms. The same will happen in this case as well too.

We can try to apply the weak Young's inequality to  $\varphi(x) = \prod |x_i|^{1-\alpha_i}$ . This would require us calculate  $\|\varphi\|_{q,w}$  to show that it is in the weak  $L_w^q$  space.

**Is fractional noise in the weak  $L^p$  space?**

Let  $\varphi(x) = \prod_{j=1}^d |x_j|^{\alpha_j-1}$ . We would like for  $\varphi \in L_w^{d/(d-\alpha)}$  because then we would apply weak young's inequality with  $q = d/(d-\alpha)$  and see that

$$\int_{\mathbb{R}^{2d}} F(x)\varphi(x-y)G(y) \leq \|\varphi\|_{d/(d-\alpha),w} \|F\|_p \|G\|_r$$

where  $1/p + 1/r = 1 + \alpha/d$ . Then we could follow the Xia Chen paper exactly to show  $\rho$  is finite.

From the definition of the weak  $L^p$  space on wiki, we need to show that for

$$K_\varphi(t) = \left| \left\{ x : \prod_{j=1}^d |x_j|^{\alpha_j-1} > t \right\} \right|$$

that there exists a  $C$  such that

$$K_\varphi(t) \leq \frac{C^p}{t^p}.$$

**Nick's Question 1.9 More about the fractional noise** The weak young's inequality says

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x)g(x-y)g(y)dx dy \right| \leq K_{p,q,r,d} \|f\|_p \|g\|_{q,w} \|h\|_r$$

w here  $p, q, r > 1$  and  $1/p + 1/q + 1/r = 2$  and

$$\|g\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |f(x)|dx$$

where  $A$  is a arbitrary Borel set of finite measure.

**Weak  $L^q$  norm for the Riesz Kernel**

In Analysis - Lieb they state that for  $f(x) = |x|^{-\lambda}$  that

$$\left\| |\cdot|^{-\lambda} \right\|_{\frac{d}{\lambda},w} = \frac{d}{d-\lambda} \left[ |\mathbb{S}^{d-1}|/d \right]^{\lambda/d} \quad \text{when } f(x) = |x|^{-\lambda}$$

and to prove this they do the following. Consider  $B_R(0)$  the ball of radius  $R$  centered at the origin. First, by using polar coordinates

$$\int_{B_R(0)} \frac{1}{|x|^\lambda} dx = \int_{\mathbb{S}^{d-1}} \int_0^R r^{-\lambda} r^{d-1} dr d\sigma(y) = |\mathbb{S}^{d-1}| \frac{R^{d-\lambda}}{d-\lambda}$$

and then using the equality

$$|B_R(0)| = R^d |B_1(0)| = R^d \frac{|\mathbb{S}^{d-1}|}{d}.$$

Next note that since  $q = d/\lambda$  then

$$\frac{1}{q'} = 1 - \lambda/d = (d-\lambda)/d$$

and so

$$|B_R(0)|^{-1/q'} = \frac{d^{1-(\lambda/d)}}{R^{d-\lambda} |\mathbb{S}^{d-1}|^{\frac{d-\lambda}{d}}}$$

and thus

$$|B_R(0)|^{-1/q'} \int_{B_R(0)} |f(x)| dx = \frac{d^{1-(\lambda/d)}}{R^{d-\lambda} |\mathbb{S}^{d-1}|^{\frac{d-\lambda}{d}}} \cdot |\mathbb{S}^{d-1}| \frac{R^{d-\lambda}}{d-\lambda} = \frac{d}{d-\lambda} \left( \frac{|\mathbb{S}^{d-1}|}{d} \right)^{\lambda/d}$$

and since this is constant in  $R$  then this means that

$$\sup_{R>0} |B_R(0)|^{-1/q'} \int_{B_R(0)} |x|^{-\lambda} dx = \frac{d}{d-\lambda} \left( \frac{|\mathbb{S}^{d-1}|}{d} \right)^{\lambda/d}$$

However I am unsure why this implies that

$$\sup_A |A|^{-1/q'} \int_A |x|^{-\lambda} dx = \frac{d}{d-\lambda} \left( \frac{|\mathbb{S}^{d-1}|}{d} \right)^{\lambda/d}.$$

Lieb is only taking the sup over all balls centered at the origin instead of over all Borel sets of finite measure.

#### Possible Weak $L^q$ norm for the fractional Kernel

We can generalize the above to the kernel  $f(x) = \prod_{j=1}^d |x_j|^{-\lambda_j}$  where  $\lambda_j \in (0, 1)$ . Above, Lieb said we can take the sup over all  $B_R(0)$ . In this example I am assuming we can take the sup over all  $[-r, r] \times \cdots [-r, r] = R_r(0)$  which are the rectangles centered at the origin in  $\mathbb{R}^d$ . I do not know if this is allowed but if so then it works.

First note that  $|R_r(0)| = r^d$  and that

$$\int_{R_r(0)} \prod_{j=1}^d |x_j|^{-\lambda_j} dx = 2^d \prod_{j=1}^d \int_0^r x_j^{-\lambda_j} dx_j = \frac{2^d r^{d-\sum_{j=1}^d \lambda_j}}{\prod_{j=1}^d (1-\lambda_j)}.$$

we then choose

$$\frac{1}{q'} = \frac{d - \sum_{j=1}^d \lambda_j}{d}$$

which implies that

$$\frac{1}{q} = 1 - \frac{1}{q'} = \frac{\sum_{j=1}^d \lambda_j}{d}$$

and finally we see that

$$|R_r(0)|^{-1/q'} \int_{R_r(0)} \prod_{j=1}^d |x_j|^{-\lambda_j} dx = r^{-(d-\sum_{j=1}^d \lambda_j)} \cdot \frac{2^d r^{d-\sum_{j=1}^d \lambda_j}}{\prod_{j=1}^d (1-\lambda_j)} = \frac{2^d}{\prod_{j=1}^d (1-\lambda_j)}$$

so the above quantity is independent of  $r$  and therefore,

$$\sup_r |R_r(0)|^{-1/q'} \int_{R_r(0)} \prod_{j=1}^d |x_j|^{-\lambda_j} dx = \frac{2^d}{\prod_{j=1}^d (1-\lambda_j)}.$$

What I need to verify is that if this implies that

$$\sup_A |A|^{-1/q'} \int_A \prod_{j=1}^d |x_j|^{-\lambda_j} dx = \frac{2^d}{\prod_{j=1}^d (1-\lambda_j)}. \quad (1.15)$$

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If it does, then we may use the fractional kernel in the paper. The required conditions on  $p, \lambda$  and  $r$  would be

$$\begin{aligned} \implies \frac{1}{p} + \frac{1}{q} + \frac{1}{r} &= 2 \\ \implies \frac{1}{p} + \frac{\sum_{j=1}^d \lambda_j}{d} + \frac{1}{r} &= 2 \\ \implies \frac{1}{p} + \frac{1}{r} &= 2 - \frac{\sum_{j=1}^d \lambda_j}{d} \end{aligned}$$

and similar to the Reisz Kernel case, if we set  $\lambda_j = 1 - \lambda'_j$  then we get

$$\frac{1}{p} + \frac{1}{r} = 1 + \frac{\sum_{j=1}^d \lambda_j}{d}$$

and this is a generalization of the line right above equation (1.27) from the Xia Chen paper.

fracnoise

**Nick's Question 1.10** [Proof that the Fractional Noise may be applied.] Here I prove that we may consider a noise with correlation function given by

$$\gamma(x) = \prod_{j=1}^d |x_j|^{-\alpha_j}.$$

The spectral density for this noise is given by

$$\varphi(x) = \prod_{j=1}^d |x_j|^{-(1-\alpha_j)}.$$

This noise satisfies all the scaling properties. The only issue is showing that  $\rho$  for this noise is finite. This can be shown by proving Lemma 1.6 from the Xia Chen paper but by applying it to the fractional noise. The exact statement we prove is the following:

**Lemma:** For any  $f, g, h$  with  $h \geq 0$ , we have the following:

$$\left( \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{|f(x+y)g(y)|}{\sqrt{h(x+y)}\sqrt{h(y)}} dy \right]^p \varphi(x) dx \right)^{1/p} \leq C \|f\|_2 \|g\|_2 \|h^{-1}\|_{pd/\alpha}$$

where  $\alpha = \sum_{j=1}^d \alpha_j$  with  $\alpha_j \in (0, 1)$  and  $\varphi(x) = \prod_{j=1}^d |x_j|^{-(1-\alpha_j)}$ .

**proof:** By observing the proof of Lemma 1.6 from the Xia Chen paper, we notice that we only need to prove equation (1.27) which says

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)G(x)\varphi(x-y)dydx \leq C \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)}. \quad (1.16)$$

where

$$F(x) = \frac{|f(x)|}{(h(x))^{p/2}} \quad \text{and} \quad G(x) = \frac{|g(x)|}{(h(x))^{p/2}}$$

If we can do this then the result follows by copying the rest of the proof in the Xia Chen paper. To prove 1.17, we use the weak version of Young's inequality. Recall that weak

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Young's inequality says

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} a(x)b(x-y)c(y)dx dy \right| \leq K_{p,q,r,d} \|a\|_p \|b\|_{q,w} \|c\|_r$$

where  $p, q, r > 1$  and  $1/p + 1/q + 1/r = 2$  and

$$\|b\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |f(x)| dx$$

where  $A$  is a arbitrary Borel set of finite measure. So to prove 1.17 we wish to be ably to apply this with

$$p = r = 2d/(d + \alpha) \quad \text{and} \quad q = d / \left( d - \sum_{j=1}^d \alpha_j \right) = d/(d - \alpha)$$

and, in addition, with

$$a = F, \quad c = G \quad \text{and} \quad b = \varphi$$

Again, by following the steps of the proof from the Xia Chen paper, we note that it suffices to show that  $\|\varphi\|_{q,w}$  is finite with  $q = d/(d - \alpha)$ .

Denote  $R_r(0) := [-r, r] \times \cdots \times [-r, r]$  to be the cube centered at the origin in  $\mathbb{R}^d$  with side length  $2r$ . With this we have that

$$\int_{R_r(0)} \prod_{j=1}^d |x_j|^{-(1-\alpha_j)} dx = 2^d \prod_{j=1}^d \int_0^r x_j^{-(1-\alpha_j)} dx_j = \frac{2^d r^{\sum_{j=1}^d \alpha_j}}{\prod_{j=1}^d \alpha_j} = \frac{2^d r^\alpha}{\prod_{j=1}^d \alpha_j}.$$

Note that

$$\frac{1}{q'} = 1 - \frac{1}{q} = 1 - \frac{d - \alpha}{d} = \frac{\alpha}{d}$$

and so

$$|R_r(0)|^{-1/q'} = (2^d r^d)^{-\alpha/d} = \frac{1}{2^{\alpha} r^\alpha}.$$

Hence,

$$|R_r(0)|^{-1/q'} \int_{R_r(0)} \prod_{j=1}^d |x_j|^{-(1-\alpha_j)} dx = \frac{2^{d-\alpha}}{\prod_{j=1}^d \alpha_j}$$

and this is independent of  $r$  so

$$\sup_{r>0} |R_r(0)|^{-1/q'} \int_{R_r(0)} \prod_{j=1}^d |x_j|^{-(1-\alpha_j)} dx = \frac{2^{d-\alpha}}{\prod_{j=1}^d \alpha_j}.$$

However, by the symmetry of  $\varphi$  we also have that

$$\sup_{r>0} |R_r(0)|^{-1/q'} \int_{R_r(0)} \prod_{j=1}^d |x_j|^{-(1-\alpha_j)} dx = \sup_A |A|^{-1/q'} \int_{R_r(0)} \prod_{j=1}^d |x_j|^{-(1-\alpha_j)} dx$$

where  $A$  denotes the Borel sets of finite measure. So finally we see that

$$\|\varphi\|_{\frac{d}{d-\alpha}, w} = \frac{2^{d-\alpha}}{\prod_{j=1}^d \alpha_j} < \infty$$

and by weak Young's inequality we have that

$$\begin{aligned} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)G(x)\varphi(x-y)dydx &\leq K \|\varphi\|_{\frac{d}{d-\alpha},w} \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)} \\ &= K \frac{2^{d-\alpha}}{\prod_{j=1}^d \alpha_j} \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)} \end{aligned}$$

which completes the proof of the Lemma.

### Nick's Question 1.11 Proof of justification of generalized fractional noise

Here I generalize the result from 1.10 to a more general Gaussian noise with correlation function

$$\gamma(x) = \prod_{i=1}^k |x_{(i)}|^{-\alpha_i}$$

where we partition  $\{1, \dots, d\}$  into  $k$  groups of size  $d_1, \dots, d_k$  and  $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ . So  $d = d_1 + \dots + d_k$  and, in addition,  $\alpha_i \in (0, d_i)$ . We define  $\alpha = \sum_{i=1}^k \alpha_i$ .

The spectral density of this noise is given by

$$\varphi(x) = \prod_{i=1}^k |x_{(i)}|^{-(d_i - \alpha_i)}$$

**Lemma:** For any  $f, g, h$  with  $h \geq 0$ , we have the following:

$$\left( \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{|f(x+y)g(y)|}{\sqrt{h(x+y)}\sqrt{h(y)}} dy \right]^p \varphi(x) dx \right)^{1/p} \leq C \|f\|_2 \|g\|_2 \|h^{-1}\|_{pd/\alpha}$$

where  $\alpha = \sum_{i=1}^k \alpha_i$  with  $k \in \{1, \dots, d\}$ ,  $\alpha_i \in (0, d_i)$  and  $\varphi(x) = \prod_{i=1}^k |x_{(i)}|^{-(d_i - \alpha_i)}$  where  $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$  is the  $i^{th}$  group of  $x = (x_1, \dots, x_d)$ .

**proof:** By observing the proof of Lemma 1.6 from the Xia Chen paper, we notice that we only need to prove equation (1.27) which says

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)G(x)\varphi(x-y)dydx \leq C \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)}. \quad (1.17) \quad \boxed{\text{fracrhofinite}}$$

where

$$F(x) = \frac{|f(x)|}{(h(x))^{p/2}} \quad \text{and} \quad G(x) = \frac{|g(x)|}{(h(x))^{p/2}}$$

If we can do this then the result follows by copying the rest of the proof in the Xia Chen paper. To prove 1.17, we use the weak version of Young's inequality. Recall that weak Young's inequality says

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} a(x)b(x-y)c(y)dx dy \right| \leq K_{p,q,r,d} \|a\|_p \|b\|_{q,w} \|c\|_r$$

where  $p, q, r > 1$  and  $1/p + 1/q + 1/r = 2$  and

$$\|b\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |f(x)| dx$$

where  $A$  is an arbitrary Borel set of finite measure. So to prove 1.17 we wish to be able to apply this with

$$p = r = 2d/(d + \alpha) \quad \text{and} \quad q = d / \left( d - \sum_{j=1}^d \alpha_j \right) = d/(d - \alpha)$$

and, in addition, with

$$a = F, \quad c = G \quad \text{and} \quad b = \varphi$$

Again, by following the steps of the proof from the Xia Chen paper, we note that it suffices to show that  $\|\varphi\|_{q,w}$  is finite with  $q = d/(d - \alpha)$ . Note that  $1/q' = 1 - 1/q = \alpha/d$ .

First we define  $A_R := A_1 \times \cdots \times A_k$  where  $A_i \subset \mathbb{R}^{d_i}$  and  $A_i = [-R, R]$  if  $d_i = 1$  and  $A_i = B_R(0)$  if  $d_i > 1$  where  $B_R(0)$  is the ball centered at the origin with radius  $R$ . With this we have that

$$\int_{A_i} |x_{(i)}|^{-(d_i - \alpha_i)} dx_{(i)} = \begin{cases} 2 \frac{R^{\alpha_i}}{\alpha_i} & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{\alpha_i}}{\alpha_i} & , d_i > 1 \end{cases} \quad (1.18) \quad \text{int}$$

where we used polar coordinated to calculate the integral when  $d_i > 1$  and  $|\mathbb{S}^{d_i-1}|$  is the surface area of the sphere in  $\mathbb{R}^{d_i}$ . Also,

$$|A_i| = \begin{cases} 2R & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{d_i}}{d_i} & , d_i > 1. \end{cases} \quad (1.19) \quad \text{leb}$$

Recall that  $1/q' = \alpha/d$ . Then by combining 1.18 and 1.19 we see that

$$\begin{aligned} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx &= \left( \prod_{i=1}^k |A_i| \right)^{-1/q'} \prod_{i=1}^k \int_{A_i} |x_{(i)}|^{-(d_i - \alpha_i)} dx_{(i)} \\ &= R^{-\alpha} \left( \prod_{i=1}^k \begin{cases} 2 & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{1}{d_i} & , d_i > 1. \end{cases} \right) \times R^{\alpha} \left( \prod_{i=1}^k \begin{cases} 2 \frac{R^{\alpha_i}}{\alpha_i} & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{\alpha_i}}{\alpha_i} & , d_i > 1 \end{cases} \right) \\ &= K \end{aligned}$$

where  $K$  is a constant independent of  $R$ . Hence we that

$$\sup_{R>0} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx = K.$$

and by the symmetry of  $\varphi$  we can in fact say that

$$\sup_B |B|^{-1/q'} \int_B \varphi(x) dx = K$$

where  $B$  denotes the Borel sets of finite measure. Hence,  $\varphi \in L_{q,w}(\mathbb{R}^d)$  with  $q = d/(d - \alpha)$  which completes the proof.

**Nick's Question 1.12 Representation (1.19) from the wave paper** I have a question about (1.19), here are my steps trying to prove it but I am unsure why the

last step is true. For notation, let  $\xi = (\xi_1, \dots, \xi_n)$  and  $\mu(d\xi) = \mu(d\xi_1) \cdots \mu(d\xi_n)$ . Also  $x = (x_1, \dots, x_n)$

$$\begin{aligned}
\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \int_{\mathbb{R}^{nd}} \mathcal{F}(\tilde{f}_n(\cdot, 0, t))(\xi_1, \dots, \xi_n) \mathcal{F}(\tilde{f}_n(\cdot, 0, t))(\xi_1, \dots, \xi_n) \mu(d\xi_1) \cdots \mu(d\xi_n) \\
&= \int_{\mathbb{R}^{nd}} \mathcal{F}(\tilde{f}_n(\cdot, 0, t))(\xi) \mathcal{F}(\tilde{f}_n(\cdot, 0, t))(\xi) \mu(d\xi) \\
&= (2\pi)^d \int_{\mathbb{R}^{dn}} [\tilde{f}_n(\cdot, 0, t) * \tilde{f}_n(\cdot, 0, t)](x) \Gamma(dx) \\
&= (2\pi)^d \int_{\mathbb{R}^{dn}} [\tilde{f}_n(\cdot, 0, t) * \tilde{f}_n(\cdot, 0, t)](x) \gamma(x) dx \\
&= (2\pi)^d \int_{\mathbb{R}^{dn}} [\tilde{f}_n(\cdot, 0, t) * \tilde{f}_n(\cdot, 0, t)](x) (K * K)(x) dx \\
&= (2\pi)^d \langle \tilde{f}_n(\cdot, 0, t) * \tilde{f}_n(\cdot, 0, t), K * K \rangle \\
&= (2\pi)^d \langle \tilde{f}_n(\cdot, 0, t) * K, \tilde{f}_n(\cdot, 0, t) * K \rangle \\
&= (2\pi)^d \int_{\mathbb{R}^{dn}} \left( [\tilde{f}_n(\cdot, 0, t) * K](x) \right)^2 dx \\
&= (2\pi)^d \int_{\mathbb{R}^{dn}} \left( \int_{\mathbb{R}^{dn}} \tilde{f}_n(y, 0, t) K(x - y) dy \right)^2 dx \\
&= (2\pi)^d \int_{\mathbb{R}^{dn}} \left( \int_{\mathbb{R}^{dn}} \frac{1}{n!} \sum_{\sigma \in \Sigma_n} f_n(y_{\sigma(1)}, \dots, y_{\sigma(n)}) K(x - y) dy \right)^2 dx \\
&= (2\pi)^d \int_{\mathbb{R}^{dn}} \left( \int_{\mathbb{R}^{dn}} \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n} G(s_1, y_{\sigma(1)}) \cdots G(s_n - s_{n-1}, y_{\sigma(n)} - y_{\sigma(n-1)}) ds K(x - y) dy \right)^2 dx
\end{aligned}$$

and therefore we have

$$\begin{aligned}
\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 &= (2\pi)^d \int_{\mathbb{R}^{dn}} \left( \int_{\mathbb{R}^{dn}} \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n} G(s_1, y_{\sigma(1)}) \cdots G(s_n - s_{n-1}, y_{\sigma(n)} - y_{\sigma(n-1)}) ds K(x - y) dy \right)^2 dx \\
&\quad (1.20) \quad \boxed{\text{rep}}
\end{aligned}$$

This is pretty much (1.19) but if you look at (1.19) you can see that

$$K(x - y) = K(x_1 - y_1) \cdots K(x_n - y_n)$$

but why does this have to be true?

**Nick's Question 1.13 Missing  $(2\pi)^{-d}$  constant and definition of Fourier transform** There are a couple places where this constant may be missing. This is because when applying Plancherel's Theorem with

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^d} f(x) e^{-ix\xi} dx$$

we have

$$\langle f, g \rangle = (2\pi)^{-d} \langle \mathcal{F}(f), \mathcal{F}(g) \rangle$$

and this  $(2\pi)^{-d}$  was ignored in this paper. If you see in 1.20, there is a  $(2\pi)^d$  in the front. I also believe that there is some unnecessary work dealing with the  $\mathcal{M}$  representation of  $\rho$ .

**First, in the Xia Chen paper**, we have

$$\Lambda_\alpha = \sup_{f \in \mathcal{F}_2} \left\{ \langle f^2 * f^2, \gamma \rangle^{1/2} + \mathcal{E}_2(f, f) \right\}$$

where

$$\mathcal{E}_2(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |x|^2 \mathcal{F}(f)(x) dx$$

and Xia Chen proves that

$$\rho = (2\pi)^{-d} \Lambda_\alpha^{(4-\alpha)/2} \quad (1.21)$$

xiarep

**Now, in the wave paper**, we introduce the  $\mathcal{M}$  functional

$$\mathcal{M}(f, \theta) = \sup_{g \in \mathcal{F}_2} \left\{ \langle g^2 * g^2, f \rangle^{1/2} - \frac{\theta}{2} \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx \right\}$$

because

$$\int_{\mathbb{R}^d} |\nabla g(x)|^2 dx = (2\pi)^{-d} \int_{\mathbb{R}^d} |x|^2 |\mathcal{F}g(x)|^2 dx$$

Why don't we just use 1.21? Or even just leave everything in terms of  $\rho$ ?

**However, in the Xia Chen Paper** he uses

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^d} f(x) e^{ix\xi} dx$$

as the definition of the Fourier transform and so 1.21 is proven for that definition. In the wave paper,

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^d} f(x) e^{-ix\xi} dx$$

is taken to be the definition. This means that 1.21 would need to be changed if we use this definition of the Fourier transform. Specifically, I think that all that needs to change is that we need to define

$$\mathcal{E}_2(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |x|^2 \mathcal{F}(f)(-x) dx.$$

But by a quick change of variables its easy to see that

$$\int_{\mathbb{R}^d} |x|^2 \mathcal{F}(f)(-x) dx = \int_{\mathbb{R}^d} |x|^2 \mathcal{F}(f)(x) dx$$

However, I think it would be more consistent if we follow Xia Chen and use his definition of the Fourier Transform.

**Nick's Question 1.14 (6.9) verification** that

For our definition of the Fourier we have

1.  $\widehat{f * g} = \hat{f} \hat{g}$
2.  $\widehat{fg} = (2\pi)^{-d} \hat{f} \hat{g}$
3.  $\langle f, g \rangle = (2\pi)^{-d} \langle \hat{f}, \hat{g} \rangle$  Verification of (6.2) for  $d = 2$ . First we note that we are only

now considering the Reisz Kernal,  $\gamma$ , which is an even function. Since  $G$  is also even then  $G * \gamma$  is even which means  $\mathcal{F}(G * \gamma)$  is even. Also we are only considering  $f$  to be non-negative and non-negative definite which implies that  $\mathcal{F}f$  will be even as well.

$$\begin{aligned}
W_2(t, \mathcal{F}(f)) &= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \int_{\mathbb{R}^d} \mu(d\xi_1) \mathcal{F}f(\xi_1) \mathcal{F}f(\xi_2) \mathcal{F}G(s_1, \cdot)(\xi_1 + \xi_2) \mathcal{F}(G(s_2 - s_1, \cdot)(\xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot)(\xi_2) \int_{\mathbb{R}^d} \mu(d\xi_1) \mathcal{F}f(\xi_1) \mathcal{F}G(s_1, \cdot)(\xi_1 + \xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot)(\xi_2) \int_{\mathbb{R}^d} d\xi_1 \mathcal{F}(f * \gamma)(\xi_1) \mathcal{F}G(s_1, \cdot)(\xi_1 + \xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot)(\xi_2) \int_{\mathbb{R}^d} d\xi_1 \mathcal{F}(f * \gamma)(\xi_1) \mathcal{F}G(s_1, \cdot)(\xi_1 + \xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot)(\xi_2) \int_{\mathbb{R}^d} d\xi_1 \mathcal{F}(f * \gamma)(\xi_1) \mathcal{F}G(s_1, \cdot)(-\xi_1 + \xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot)(\xi_2) [\mathcal{F}(f * \gamma) * \mathcal{F}G(s_1, \cdot)](\xi_2) \\
&= (2\pi)^{-d} \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} d\xi_2 \mathcal{F}(f * \gamma)(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot)(\xi_2) \mathcal{F}[(f * \gamma)G(s_1, \cdot)](\xi_2) \\
&= (2\pi)^{-d} \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} d\xi_2 \mathcal{F}[(f * \gamma) * G(s_2 - s_1, \cdot)](\xi_2) \mathcal{F}[(f * \gamma)G(s_1, \cdot)](\xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} dx_2 [(f * \gamma) * G(s_2 - s_1, \cdot)](x_2) [(f * \gamma)G(s_1, \cdot)](x_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} dx_2 (f * \gamma)(x_2) G(s_1, x_2) \int_{\mathbb{R}^d} dx_1 (f * \gamma)(x_1) G(s_2 - s_1, x_2 - x_1) \\
&= \int_{[0,t]^2 <} ds \iint_{\mathbb{R}^{2d}} dx_2 dx_1 (f * \gamma)(x_1) G(s_2 - s_1, x_2 - x_1) (f * \gamma)(x_2) G(s_1, x_2) \\
&= \int_{[0,t]^2 <} ds \iint_{\mathbb{R}^{2d}} dx_2 dx_1 (f * \gamma)(x_2) G(s_2 - s_1, x_2 - x_1) (f * \gamma)(x_1) G(s_1, x_1)
\end{aligned}$$

and the last line follows from a substitution  $x'_1 = x_2$  and  $x'_2 = x_1$  and the fact that  $G(t, \cdot)$  is even.

newrhoinfo

**Nick's Question 1.15 The issue with  $\nu \neq 2$**  I believe that for the fractional paper we may want to redefine  $\rho$  in the following way

$$\rho_{new} = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx) \quad (1.22) \quad \text{newrho}$$

This way we can still use equation (6.15) from the wave paper for general  $\nu$ . There are two things we need to verify.

newrhlim

1.  $\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{1}{(n!)^2} T_n \right) \rightarrow \log(\rho_{new})$  where  $T_n$  is defined below.
2. (6.15) from the wave paper holds.

From reading the Xia Chen paper I think that the (6.15) equation will work but I have to still verify the details. I also need to verify 1.

Doing this may need us to keep all of our moments asymptotic in terms of  $\rho$  and not  $\mathcal{M}_a$  and if this is done correctly, all information about  $\nu$  should be completely contained in  $\rho$ . In other words,  $\nu$  should not appear anywhere other than in the definition of  $\nu$ .

The reason why I think this can work is that if we follow the steps starting right below equation (3.2) in the Xia Chen paper but define  $Q(\lambda) = (1 + \frac{\nu}{2}|\lambda|^a)^{-1}$  then it seems like everything still works and this would lead us to define  $\rho$  like I did in 1.22.

I need to still verify everything here.

**Regarding 1.**

Using fact that  $\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{1}{(n!)^2} T'_n \right) \rightarrow \log(\rho)$  where  $T'_n$  is  $T_n$  with  $\nu = 2$  and also using the fact that  $T_n = \left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n$  then we can see that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{1}{(n!)^2} T_n \right) = \log \left[ \left( \frac{\nu}{2} \right)^{-\alpha/a} \right] + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{1}{(n!)^2} T'_n \right) = \log \left[ \left( \frac{\nu}{2} \right)^{-\alpha/a} \right] + \log(\rho)$$

which proves that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{1}{(n!)^2} T_n \right) = \log \left[ \left( \frac{\nu}{2} \right)^{-\alpha/a} \rho \right]$$

so now we need to show that  $\left(\frac{\nu}{2}\right)^{-\alpha/a} \rho = \rho_{new}$ .

**The following may give hope that  $\left(\frac{\nu}{2}\right)^{-\alpha/a} \rho = \rho_{new}$  is in fact true.**

$$\begin{aligned} \rho_{new} &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx) \\ &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \left(\frac{\nu}{2}\right)^{(d-\alpha)/a} \varphi(x) \left(\frac{\nu}{2}\right)^{-d/a} dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \varphi(x) dx \end{aligned}$$

Now define  $f^*(x) = \left(\frac{\nu}{2}\right)^{\frac{-d}{2a}} f\left(\left(\frac{\nu}{2}\right)^{-1/a}x\right)$ . Then with this we see that

$$\rho_{new} = \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f^*(x+y)f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx$$

and note that

$$\int_{\mathbb{R}^d} f(x)^2 dx = \int_{\mathbb{R}^d} f^*(x)^2 dx$$

and thus

$$\left\{ f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1 \right\} = \left\{ f \in L^2(\mathbb{R}^d) : \|f^*\|_2 = 1 \right\}$$

so we can say that

$$\sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \varphi(x) dx$$

$$\begin{aligned}
&= \sup_{\|f^*\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \varphi(x) dx \\
&= \sup_{\|f^*\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx
\end{aligned}$$

and this leads me to believe that  $\left(\frac{\nu}{2}\right)^{-\alpha/a} \rho = \rho_{new}$ .

**Variational Representation:**  $\rho_{new} = \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}$

**proof:** We have that

$$\begin{aligned}
\rho_{new} &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \rho \\
&= \left(\frac{\nu}{2}\right)^{-\alpha/a} \Lambda_\alpha^{\frac{2a-\alpha}{a}} \\
&= \left(\frac{\nu}{2}\right)^{-\alpha/a} \left(2^{\frac{-\alpha}{2a-\alpha}} \mathcal{M}_a\right)^{\frac{2a-\alpha}{a}} \\
&= \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}
\end{aligned}$$

and the reduces to the wave equation when  $\nu = 2$  and  $a = 2$ .

#### Nick's Question 1.16 comparing our Corrolary 1.8 to [13] Theorem 1.1

Notation and variables vary from paper to paper so I will restate all the necessary equations we need. Lets assume that

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-ix\xi} f(x) dx.$$

First we have Theorem 1.1 and the notation from [13]. Here we have

|               | Laplace  |       | Noise            |                 |                                    | Variational Const.                    |
|---------------|----------|-------|------------------|-----------------|------------------------------------|---------------------------------------|
| [13]          | $\alpha$ | 2     | $\beta_0$ (time) | $\beta$ (space) | $\gamma(\cdot) =  \cdot ^{-\beta}$ | <b>M</b>                              |
| Current paper | $a$      | $\nu$ | 0                | $\alpha$        | $\gamma(\cdot)$                    | $\mathcal{M}_a( \cdot ^{-\alpha}, 2)$ |

**Theorem 1.1.**

$$\lim_{t \rightarrow \infty} t^{-\frac{2\alpha-\beta-\alpha\beta_0}{\alpha-\beta}} \log \|u^\rho(t, x)\|_p = (p - \rho)^{\frac{\alpha}{\alpha-\beta}} \mathbf{M}(\alpha, \beta_0, d, \gamma)$$

and in this theorem

$$\mathbf{M}(\alpha, \beta_0, d, \gamma)$$

$$= \sup_g \left\{ \frac{1}{2} \int_0^1 \int_0^1 \int_{\mathbb{R}^{2d}} \frac{\gamma(x-y)}{|r-s|^{\beta_0}} g^2(s, x) g^2(r, y) dx dy dr ds - (2\pi)^{-d} \int_0^1 \int_{\mathbb{R}^d} |\xi|^\alpha |\hat{g}(s, \xi)|^2 d\xi ds \right\}.$$

and also it is stated that we have the following scaling property

$$\mathbf{M}(\alpha, \beta_0, d, \theta\gamma) = \theta^{\frac{\alpha}{\alpha-\beta}} \mathbf{M}(\alpha, \beta_0, d, \gamma)$$

Now we need some results from [2]. Here we have that

|               | Laplace |       | Noise             |                  |                                     | Variational Const.                    |
|---------------|---------|-------|-------------------|------------------|-------------------------------------|---------------------------------------|
| [2]           | 2       | 2     | $\alpha_0$ (time) | $\alpha$ (space) | $\gamma(\cdot) =  \cdot ^{-\alpha}$ | $\mathcal{E}$                         |
| Current paper | $a$     | $\nu$ | 0                 | $\alpha$         | $\gamma(\cdot)$                     | $\mathcal{M}_a( \cdot ^{-\alpha}, 2)$ |

**Lemma 1.2.** *Lemma A.2 [2]*

$$\mathcal{E}(\alpha_0, d, \gamma) = \frac{2-\alpha}{2} 2^{\alpha/(2-\alpha)} \left( \frac{4-\alpha}{4} \right)^{-\frac{4-\alpha}{2-\alpha}} M(\alpha_0, d, \gamma)^{\frac{4-\alpha}{2-\alpha}}$$

and in this lemma we have

$$\begin{aligned} \mathcal{E}(\alpha_0, d, \gamma) &= \sup_g \left\{ \int_0^1 \int_0^1 \int_{\mathbb{R}^{2d}} \frac{\gamma(x-y)}{|r-s|^{\alpha_0}} g^2(s, x) g^2(r, y) dx dy dr ds - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right\} \\ &= \sup_g \left\{ \int_0^1 \int_0^1 \int_{\mathbb{R}^{2d}} \frac{\gamma(x-y)}{|r-s|^{\alpha_0}} g^2(s, x) g^2(r, y) dx dy dr ds - \frac{(2\pi)^{-d}}{2} \int_0^1 \int_{\mathbb{R}^d} |\xi|^2 |\hat{g}(s, \xi)|^2 d\xi ds \right\} \end{aligned}$$

and

$$\begin{aligned} M(\alpha_0, d, \gamma) &= \sup_g \left\{ \left( \int_0^1 \int_0^1 \int_{\mathbb{R}^{2d}} \frac{\gamma(x-y)}{|r-s|^{\alpha_0}} g^2(s, x) g^2(r, y) dx dy dr ds \right)^{1/2} - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right\} \\ &= \sup_g \left\{ \left( \int_0^1 \int_0^1 \int_{\mathbb{R}^{2d}} \frac{\gamma(x-y)}{|r-s|^{\alpha_0}} g^2(s, x) g^2(r, y) dx dy dr ds \right)^{1/2} - \frac{(2\pi)^{-d}}{2} \int_0^1 \int_{\mathbb{R}^d} |\xi|^2 |\hat{g}(s, \xi)|^2 d\xi ds \right\} \end{aligned}$$

We know want to relate this to  $\mathcal{M}_a$  from our paper. From this point on we will use the notation from our paper, ie, the Laplace exponent will be  $a$  and the Riesz exponent will be  $\alpha$ . Note that the above Lemma is only for  $a = 2$  so lets just focus on that for now. Also we want to consider the time independent case so we need  $\alpha_0 = \beta_0 = 0$ . Lastly, in our paper we set  $\theta = 1$  and  $\nu = 2$ . We have that

$$M(0, d, \gamma) = \mathcal{M}_{2,d}(\gamma, 1)$$

and so this implies that

$$\mathcal{E}(0, d, \gamma) = \frac{2-\alpha}{2} 2^{\alpha/(2-\alpha)} \left( \frac{4-\alpha}{4} \right)^{-\frac{4-\alpha}{2-\alpha}} \mathcal{M}_{2,d}(\gamma, 1)^{\frac{4-\alpha}{2-\alpha}}.$$

In addition we have

$$\frac{1}{2} \mathbf{M}(2, 0, d, 4\gamma) = \mathcal{E}(0, d, \gamma)$$

and by the scaling property we have

$$\frac{1}{2} \mathbf{M}(2, 0, d, 4\gamma) = \frac{1}{2} (4)^{\frac{2}{2-\alpha}} \mathbf{M}(2, 0, d, \gamma) = 2^{\frac{2+\alpha}{2-\alpha}} \mathbf{M}(2, 0, d, \gamma).$$

Now we can say that

$$2^{\frac{2+\alpha}{2-\alpha}} \mathbf{M}(2, 0, d, \gamma) = \frac{2-\alpha}{2} 2^{\alpha/(2-\alpha)} \left( \frac{4-\alpha}{4} \right)^{-\frac{4-\alpha}{2-\alpha}} \mathcal{M}_{2,d}(\gamma, 1)^{\frac{4-\alpha}{2-\alpha}}$$

and finally

$$\begin{aligned} \mathbf{M}(2, 0, d, \gamma) &= 2^{\frac{-2-\alpha}{2-\alpha}} \frac{2-\alpha}{2} 2^{\alpha/(2-\alpha)} \left( \frac{4-\alpha}{4} \right)^{-\frac{4-\alpha}{2-\alpha}} \mathcal{M}_{2,d}(\gamma, 1)^{\frac{4-\alpha}{2-\alpha}} \\ &= (2-\alpha)(4-\alpha)^{-\frac{4-\alpha}{2-\alpha}} 2^{\frac{4-\alpha}{2-\alpha}} \mathcal{M}_{2,d}(\gamma, 1)^{\frac{4-\alpha}{2-\alpha}} \end{aligned}$$

and so by the above theorem we should have

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{2}{2-\alpha}} (2-\alpha)(4-\alpha)^{-\frac{4-\alpha}{2-\alpha}} 2^{\frac{4-\alpha}{2-\alpha}} \mathcal{M}_{2,d}(\gamma, 1)^{\frac{4-\alpha}{2-\alpha}}$$

««i HEAD and by our corollary 1.8, we have that

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{2}{2-\alpha}} (2-\alpha)(4-\alpha)^{-\frac{4-\alpha}{2-\alpha}} 2^{\frac{4-\alpha}{2-\alpha}} \mathcal{M}_{2,d}(\gamma, 1)^{\frac{4-\alpha}{2-\alpha}}$$

so the two results match.

**Nick's Question 1.17 Lemma A.3 from [10]** In order to prove Lemma A.2 [2], we need to prove a version of Lemma A.1 *ibid.* for general "a" and according to [2], we need to prove Lemma 5.7 [11]. However this lemma does not exist in that paper and what I believe Xia Chen means is (A.17) from [11]. This lemma says that for  $0 \leq p \leq \min 2, d$ , there exists a  $C > 0$  such that

$$\int_{\mathbb{R}^d} \frac{f^2(x)}{|x|^p} dx \leq C \|f\|_2^{2-p} \|\nabla f\|_2^p \quad f \in W^{1,2}(\mathbb{R}^d). \quad (1.23) \quad \text{RieszIneq1}$$

However, I believe that we only really care about functions  $f \in \mathcal{A}_d$  where

$$\mathcal{A}_d = \left\{ g(s, x) \in W^{1,2}(\mathbb{R}^d) : \int_{\mathbb{R}^d} g(s, x)^2 dx = 1 \quad \forall s \in [0, 1], \text{ and } \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds < \infty \right\}.$$

The proof of this is not given in [11] and it claims that it is proven in Lemma 7.2 of [10] which again does not exist and I believe that Xia Chen means Lemma (A.3) [10]. However this lemma only says that (1.23) is true only for  $d/2 \leq p \leq \min 2, d$  so there is a little discrepancy here.

In the first line of the proof in [10], it says consider  $f \in W^{1,2}(\mathbb{R}^d)$  with  $\|f\|_2 = 1$  and let

$$\int_{\mathbb{R}^d} \frac{f^2(x)}{|x|^p} dx = C_f \|\nabla f\|_2^p.$$

How do we know the integral on the left hand side is finite? I believe it is because we know that

$$M(\lambda) = \sup_{\|g\|_{L^2(\mathbb{R}^d)}=1} \left\{ \lambda \int_{\mathbb{R}^d} \frac{g^2(x)}{|x|^p} dx - \frac{1}{2} \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx \right\} < \infty.$$

This is stated in equation (A.2) [10]. Now if you continue following the proof in [10], they show that

$$M(1) \geq \frac{2-p}{2} C_f^{2/(2-p)} p^{p/(2-p)}$$

which I believe shows that

$$\sup_{f \in W^{1,2}(\mathbb{R}^d), \|f\|_2=1} C_f =: C < \infty$$

which in return implies that

$$\int_{\mathbb{R}^d} \frac{f^2(x)}{|x|^p} dx \leq C \|\nabla f\|_2^p$$

which shows that (1.23) is true for  $f$  such that  $\|f\|_2 = 1$ . To show this is true for general  $f \in W^{1,2}(\mathbb{R}^d)$ , take an arbitrary  $f \in W^{1,2}(\mathbb{R}^d)$  then  $h := f / \|f\|_2$  satisfies  $\|h\|_2 = 1$  and so (1.23) implies that

$$\int_{\mathbb{R}^d} \frac{h^2(x)}{|x|^p} dx \leq C \|\nabla h\|_2^p$$

which in return implies that

$$\frac{1}{\|f\|_2^2} \int_{\mathbb{R}^d} \frac{f^2(x)}{|x|^p} dx \leq C \|f\|_2^{-p} \|\nabla f\|_2^p$$

which gives (1.23).

**Remark 1.3.** Mismatches of the results in the reference happen often because the final version of the paper may have undergone some major revision.

**Remark 1.4.** (1.23) is called *interpolation inequality*:

[https://en.wikipedia.org/wiki/Interpolation\\_inequality](https://en.wikipedia.org/wiki/Interpolation_inequality)

Please check the Wikipage, where you can find how to generalize this to some  $H$  spaces.

**Nick's Question 1.18 Concern about**  $\int_{\mathbb{R}^d} |\nabla f(x)|^2 dx = \int_{\mathbb{R}^d} |x|^2 |\mathcal{F}f(x)|^2 dx$

Just for simplicity lets suppose that  $d = 1$ , so we want to show that

$$\int_{\mathbb{R}^d} |g'(x)|^2 dx = \int_{\mathbb{R}^d} |x|^2 |\mathcal{F}f(x)|^2 dx.$$

By Placherel's Theorem, the left hand side becomes

$$\int_{\mathbb{R}^d} |g'(x)|^2 dx = (2\pi)^{-d} \int_{\mathbb{R}^d} |\mathcal{F}(f')(x)|^2 dx.$$

Now we want to be able to say that

$$|\mathcal{F}f'(x)| = |x| |\mathcal{F}f(x)| \tag{1.24}$$

Ff'

but according to [wikipedia](#), this is only true if  $f$  is absolutely continuous. It can also be shown to be true if  $\lim_{|x| \rightarrow \infty} f(x) = 0$ . However I am unsure if either of these are true. The second situation is not the case here because all we are assuming is that  $f \in W^{1,2}(\mathbb{R}^d)$  which does not imply that  $\lim_{|x| \rightarrow \infty} f(x) = 0$ .

As far as absolute continuity, I can only find that functions  $f \in W^{1,p}(0,1)$  are absolutely continuous. As far as function  $f \in W^{1,p}(\mathbb{R}^d)$  I have only been able to find that  $f(x') = g_C(x')$  almost surely and  $g_C$  is absolutely continuous and  $x' \in C$  where  $C$  is any curve in  $\mathbb{R}^d$ . This result is found [here](#). However I am unsure if this is good enough to say (1.24)

**Remark 1.5.** This question is related to the Fourier multiplier. For (1.24), Theorem 7.4 (c) on p. 184 of Rudin's *Functional Analysis* says that (1.24) is true for all Schwartz test functions. Then through a standard dense argument, this relation still holds provided either of the two integrals right above (1.24) is finite.

**Remark 1.6.** In principle, the generalization of Lemma A.2 of [14] should be able to achieved by replacing everywhere  $\int_{\mathbb{R}^d} |\nabla f(x)|^2 dx$  by

$$(2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}f(\xi)|^2 d\xi.$$

You may first assume all functions are Schwartz test functions. Then by integrability, we can extend the desired results to larger class of functions, such as  $\mathcal{F}_a$  in our paper.

**Nick's Question 1.19 Generalizing Lemma (A.3) [10]** Take  $f \in \mathcal{F}_a$ , ie,  $\|f\|_2 = 1$  and  $\mathcal{E}(f, f) < \infty$ . Choose  $C_f$  such that

$$\int_{\mathbb{R}^d} \frac{f^2(x)}{|x|^\alpha} dx = C_f \left( \int_{\mathbb{R}^d} |x|^a |\mathcal{F}f(x)|^2 dx \right)^{\alpha/2} =: C_f I_a(f)^\alpha.$$

Did you prove the above inequality? And why can you find such  $f$  so that this inequality becomes an equality?

Let  $t > 0$  and define

$$g(x) = t^{d/2} f(tx).$$

Then  $\|g\|_2 = 1$  and in addition  $I_a(g) = t^{a/2} I_a(f)$ . With this we see that

$$\int_{\mathbb{R}^d} \frac{g^2(x)}{|x|^\alpha} dx = t^\alpha \int_{\mathbb{R}^d} \frac{f^2(x)}{|x|^\alpha} dx = t^\alpha C_f I_a(f)^\alpha = C_f t^{(2-a)\alpha/2} I_a(g)^\alpha$$

Define

$$M(\lambda) := \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \left\{ \lambda \int_{\mathbb{R}^d} \frac{h^2(x)}{|x|^\alpha} dx - \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} |x|^a |\mathcal{F}h(x)|^2 dx \right\}.$$

It can be shown that

$$M(\lambda) = \lambda^{2/(2-p)} M(1).$$

We now see that

$$\begin{aligned} M(1) &\geq C_f t^{(2-a)\alpha/2} I_a(g)^\alpha - \frac{1}{(2\pi)^2} I_a(g)^2 \\ &= C_f t^\alpha I_a(f)^\alpha - \frac{1}{(2\pi)^d} t^a I_a(f)^2 \\ &= C_f \underbrace{(t I_a(f))^\alpha}_{(t I_a(f))^\alpha} - \frac{1}{(2\pi)^d} \underbrace{(t^{a/2} I_a(f))^2}_{(t^{a/2} I_a(f))^2} \end{aligned}$$

This is an issue because we need the items inside the parenthesis of the underlines terms to be the same in order to continue with the proof.

**Issue with  $a \neq 2$  case with Lemma A.2 [2]**

To try and fix the above issue we can try to search of a different function  $g$ . We can guess that  $g$  takes the form

$$g(x) = t^p f(t^q x).$$

Forsure sure we need that

$$\|g\|_2 = 1$$

and this implies that

$$2p = dq. \tag{1.25} \quad \boxed{\text{condition1}}$$

Next we want to calculate both

$$\int_{\mathbb{R}^d} \frac{g^2(x)}{x^\alpha} dx \quad \text{and} \quad I_a(g).$$

We can show the following:

1.

$$\left( \int_{\mathbb{R}^d} |x|^a |\mathcal{F}g(x)|^2 dx \right)^{1/2} = t^{p+(\frac{a-d}{2})q} \left( \int_{\mathbb{R}^d} |x|^a |\mathcal{F}f(x)|^2 dx \right)^{1/2}$$

2.

$$\int_{\mathbb{R}^d} \frac{g^2(x)}{x^\alpha} dx = t^{2p+(\alpha-d)q} \int_{\mathbb{R}^d} \frac{f^2(x)}{|x|^\alpha} dx$$

3.

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{g^2(x)}{|x|^\alpha} dx &= t^{2p+(\alpha-d)q-\alpha p - (\frac{a-d}{2})\alpha q} C_f \left( \int_{\mathbb{R}^d} |x|^a |\mathcal{F}f(x)|^2 dx \right)^{\alpha/2} \\ &= t^{(\frac{2\alpha-a\alpha}{2})q} C_f \left( \int_{\mathbb{R}^d} |x|^a |\mathcal{F}f(x)|^2 dx \right)^{\alpha/2} \end{aligned} \quad (1.26) \quad \text{condition4}$$

where in (1.26), we used (1.25).

Note that when  $a = 2$  and when  $p = d/2$  and  $q = 1$  then everything reduces back to what is presented in the proof of Lemma A.3 [10].

Our goal is to reproduce the third centered equation from the proof lemma A.3 [10] which says that

$$M(1) \geq C_f t^\alpha I_a(f)^\alpha - \frac{1}{2} t^2 I_a(f)^2 \quad (1.27) \quad \text{step1}$$

which is for the case when  $a = 2$  and  $p = d/2$  and  $q = 1$ .

In the general case, we get that

$$M(1) \geq \int_{\mathbb{R}^d} \frac{g^2(x)}{|x|^\alpha} dx - \frac{(2\pi)^{-d}}{2} \int_{\mathbb{R}^d} |x|^a |\mathcal{F}g(x)|^2 dx \quad (1.28)$$

$$t^{(\lfloor \frac{2-d}{2} \rfloor q + p)\alpha} C_f I_a(f)^\alpha - \frac{(2\pi)^{-d}}{2} t^{(\lfloor \frac{a-d}{2} \rfloor q + p)^2} I_a(f)^2. \quad (1.29) \quad \text{step2}$$

Note that again if we restrict ourselves to the case of  $a = 2$  and  $p = d/2$  and  $q = 1$  then we recover the equations from Lemma A.3 [10].

Note that by (1.27), that if we inspect (1.29) then we would want for

$$\left( \left\lfloor \frac{2-d}{2} \right\rfloor q + p \right) = \left( \left\lfloor \frac{a-d}{2} \right\rfloor q + p \right)$$

however this can only happen in two cases.

1.  $a = 2$

2.  $a \neq 2$  and  $q = 0$ .

So it seem that proof technique of Lemma A.2 [2] may work for the  $a = 2$  case.

**Remark 1.7.** Can you first download the paper [10] and name it ‘ChenAOP12.pdf’ and place it under the directory refs? I am not aware of this paper.

## 2 Proof Sketches

**Nick's Question 2.1** A mild solution for

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[ \sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2] \end{cases}$$

must satisfy the following integral equation

$$u(t, x) = 1 + \int_0^t \left( \int_{\mathbb{R}^d} u(t-s, x-y) Y(s, y) W(dy) \right) ds$$

an through a Piccard Iteration Scheme we can see that the solution should be given by

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)) \quad (2.1) \quad \text{sol}$$

where

$$\begin{aligned} f_k(x_1, \dots, x_k, x, t) &= \int_0^t \int_0^{t_k} \cdots \int_0^{t_2} Y(t-t_k, x-x_k) \cdots Y(t_2-t_1, x_2-x_1) dt_1 \cdots dt_n \\ &= \int_0^t \int_0^{t_k} \cdots \int_0^{t_2} Y(t_1, x-x_k) \cdots Y(t_k-t_{k-1}, x_2-x_1) dt_1 \cdots dt_n \end{aligned} \quad (2.2) \quad \text{solker}$$

and also

$$\mathcal{F}f_k(\cdot, x, t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0, t]^{n+1}} \prod_{k=1}^n \overline{\mathcal{F}Y(t_{t_k+1} - t_k, \cdot)} \left( \sum_{j=1}^k \xi_j \right) dt \quad (2.4) \quad \text{fsolker}$$

where  $t_{n+1} = t$

**Nick's Question 2.2 Greens Function Scaling Property** For and  $c > 0$  we have

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}Y(t, \cdot)(t^{b/a} \xi). \quad (2.5) \quad \text{fscale}$$

which is shown above.

**Nick's Question 2.3  $\mathcal{H}^{\otimes n}$  Norm Scaling Property** Suppose that  $\mu(d\xi) = \varphi(\xi)d\xi$  and  $\varphi$  satisfies  $\varphi(cx) = c^{-(d-\alpha)}$  for some  $\alpha \in (0, d)$  and for any  $c > 0$ . Also recall that

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}Y(t, \cdot)(t^{b/a} \xi).$$

Then

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \quad (2.6) \quad \text{nscale}$$

**Proof:**

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = \int_{\mathbb{R}^{nd}} \mathcal{F} \tilde{f}_n(\cdot, 0, t)^2(\vec{\xi}) \varphi(\vec{\xi}) d\vec{\xi}$$

where  $\varphi(\vec{\xi}) = \varphi(\xi_1) \cdots \varphi(\xi_n)$

$$\begin{aligned}
& \int_{\mathbb{R}^{nd}} \mathcal{F} \tilde{f}_n(\cdot, 0, t)^2 \varphi(\vec{\xi}) d\vec{\xi} = \int_{\mathbb{R}^{nd}} \left( \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^{n <}} \overline{\mathcal{F}Y(t - t_n, \cdot) \left( \sum_{i=1}^n \xi_i \right) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{2n(b+r-1)} \int_{\mathbb{R}^{nd}} \left( \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^{n <}} \mathcal{F}Y \left( 1 - \frac{t_n}{t}, \cdot \right) (t^{b/a} \sum_{i=1}^n \xi_i) \cdots \mathcal{F}Y \left( \frac{t_2}{t} - \frac{t_1}{t}, \cdot \right) (t^{b/a} \xi_1) \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left( \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^{n <}} \overline{\mathcal{F}Y(1 - t_n, \cdot) (t^{b/a} \sum_{i=1}^n \xi_i) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (t^{b/a} \xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left( \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^{n <}} \overline{\mathcal{F}Y(1 - t_n, \cdot) \left( \sum_{i=1}^n \xi_i \right) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (\xi_1)} \right)^2 \varphi(t^{-b/a} \vec{\xi}) t^{-ndb/a} d\vec{\xi} \\
& = t^{2n(b+r-1)+2n-nb\alpha/a} \int_{\mathbb{R}^{nd}} \left( \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^{n <}} \overline{\mathcal{F}Y(1 - t_n, \cdot) \left( \sum_{i=1}^n \xi_i \right) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{2n(b+r-1)+2n-nb\alpha/a} \int_{\mathbb{R}^{nd}} \mathcal{F} \tilde{f}_n(\cdot, 0, 1) (\vec{\xi})^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{[2(b+r-1)+2-nb\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \\
& = t^{[2(b+r)-b\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2
\end{aligned}$$

**Nick's Question 2.4 Lemma 2.3** We define

$$\mathcal{F}_a = \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\}$$

and then we also define

$$\mathcal{M}_a(f, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * f, g^2 \rangle^{1/2} - \frac{\theta}{2} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi \right\} \quad (2.7) \quad \text{var}$$

Now we define  $\tilde{g}(x) = B^{-d/2} g(x/B)$ . Its easy to show that  $\mathcal{F}\tilde{g}(\xi) = B^{d/2} \mathcal{F}g(B\xi)$ . By a change of variables, we can then show that

$$\int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}\tilde{g}(\xi)|^2 d\xi = B^{-a} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi$$

Next, by applying the scaling property of  $f$ ,  $f(cx) = c^{-\alpha} f(x)$ , we can show by again using a change of variables that

$$\begin{aligned}
(\tilde{g}^2 * f)(x) &= B^{-d} \int_{\mathbb{R}^d} \tilde{g}^2(y/B) f(x - y) dy \\
&= \int_{\mathbb{R}^d} g^2(u) f(x - uB) du \\
&= B^{-\alpha} \int_{\mathbb{R}^d} g^2(u) f\left(\frac{x}{B} - u\right) du \\
&= B^{-\alpha} (g^2 * f)(x/B)
\end{aligned}$$

This can be used to show that

$$\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{-\alpha} \langle g^2 * f, g^2 \rangle$$

Now we can rewrite (2.7) as the following

$$\mathcal{M}_a(\Theta f, \theta) = \frac{1}{\theta} \sup_{g \in \mathcal{F}_a} \left\{ \frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} \langle \tilde{g}^2 * f, \tilde{g}^2 \rangle^{1/2} - \frac{B^a}{2} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}\tilde{g}(\xi)|^2 d\xi \right\}$$

and as before, in order to preserve a scaling property, we want to choose  $B$  such that

$$\frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} = B^a$$

and solving for  $B$  we get

$$B = \left( \frac{\sqrt{\Theta}}{\theta} \right)^{\frac{2}{2a-\alpha}}$$

and after plugging this  $B$  in we get that

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(f, 1)$$

Some definitions for solvability (details in my notebook)

$$\begin{aligned} L_n(y_1, \dots, y_n) &= n! \int_0^\infty e^{-t} \tilde{f}_n(y_1, \dots, y_n, 0, t) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0, t]^n} \prod_{k=1}^n Y(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) ds dt \end{aligned}$$

where  $s_0 = y_{\sigma(0)} = 0$ .

$$\begin{aligned} H_n(t, \mathbf{x}) &= n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \tilde{f}_n(y_1, \dots, y_n, 0, t) d\mathbf{y} \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n Y(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\mathbf{y} ds \end{aligned}$$

$$L(x) = \int_0^\infty e^{-t} Y(t, x) dt \quad \text{and} \quad \mathcal{F}L(\xi) = \int_0^\infty e^{-t} \mathcal{F}Y(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}$$

$$L_n(y_1, \dots, y_n) = \sum_{\sigma \in \Sigma_n} L(y_{\sigma(1)}) L(y_{\sigma(2)} - y_{\sigma(1)}) \cdots L(y_{\sigma(n)} - y_{\sigma(n-1)})$$

$$\mathcal{F}L_n(\xi_1, \dots, \xi_n) = \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a}$$

We the two representations for  $T_n$

$$T_n = \int_{(\mathbb{R}^d)^n} \left[ \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) = \left( \frac{\nu}{2} \right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[ \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi})$$

We have the two possible representations for  $C'_\mu$ .

$$C'_\mu = \int_{(\mathbb{R}^d)} \left( \frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) = \left( \frac{\nu}{2} \right)^{-\alpha/a} \int_{(\mathbb{R}^d)} \left( \frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi)$$

**Nick's Question 2.5 Lemma 3.3, 3.4, and 3.5** Same proof as in wave paper. Have all details in my notebook.

**Nick's Question 2.6 Lemma 3.6** We have that

$$C'_\mu < \infty \quad \text{and} \quad T_n < (n!)^2 (C'_\mu)^n$$

**proof:** By the same spherical substitution in the wave paper we see that

$$\int_{(\mathbb{R}^d)} \left( \frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi) = \left( \int_{S^{d-1}} \varphi(\gamma) d\sigma(\gamma) \right) \left( \int_0^\infty \frac{\rho^{\alpha-1}}{(1 + \rho^a)^2} d\rho \right)$$

The first integral is shown to be finite in the wave paper. The second integral is finite if

$$2a - \alpha + 1 > 1 \tag{2.8}$$

alpha2

The same arguments from the wave paper show that

$$\begin{aligned} T_n &\leq n! \int_{(\mathbb{R}^d)^n} \sum_{\sigma \in \Sigma_n} \left[ \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &= (n!)^2 \int_{(\mathbb{R}^d)^n} \left[ \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &\leq (n!)^2 \left[ \sup_{\eta \in \mathbb{R}^d} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left( \frac{1}{1 + \frac{\nu}{2} |\xi + \eta|^a} \right)^2 \mu(d\xi) \right]^n \\ &= (n!)^2 \left[ \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left( \frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) \right]^n \\ &= (n!)^2 (C'_\mu)^n \end{aligned}$$

**Nick's Question 2.7 Lemma 3.7**

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^n dt \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} }{(n!)^2} T_n \leq 2^{n[2(b+r) - \frac{b\alpha}{a}]} (C'_\mu)^n$$

**Proof:** We first derive a scaling property for  $\mathcal{F}\tilde{f}_n$  where  $\tilde{f}_n$  is the kernel of the  $n^{th}$  Wiener integral in the chaos expansion of the solution. Let  $c > 0$ .

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$$

so it suffices to just look at the  $\mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$  terms individually.

$$\begin{aligned}\mathcal{F}f(\cdot, 0, ct)(\xi_1, \dots, \xi_n) &= \int_{[0, ct]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}Y(t_{k+1} - t_k, \cdot) \left( \sum_{j=1}^k \xi_j \right)} d\vec{t} \\ &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}Y(c(t_{k+1} - t_k), \cdot) \left( \sum_{j=1}^k \xi_j \right)} c^n d\vec{t}\end{aligned}$$

where we applied the substitution  $t'_i = t_i/c$ . Now by applying the scaling property of  $\mathcal{F}Y$  we see that

$$\begin{aligned}\int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}Y(c(t_{k+1} - t_k), \cdot) \left( \sum_{j=1}^k \xi_j \right)} c^n d\vec{t} &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{c^{b+r-1} \mathcal{F}Y(t_{k+1} - t_k, \cdot) \left( c^{b/a} \sum_{j=1}^k \xi_j \right)} c^n d\vec{t} \\ &= c^{n(b+r)} \mathcal{F}f(\cdot, 0, t)(c^{b/a} \xi_1, \dots, c^{b/a} \xi_n)\end{aligned}$$

Now by using this scaling property we see that

$$\begin{aligned}\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= \int_0^\infty e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, 2t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) 2dt \\ &= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, t)(2^{b/a} \xi_1, \dots, 2^{b/a} \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, t)(\xi_1, \dots, \xi_n) \right|^2 2^{\frac{-nbd}{a}} 2^{\frac{nb(d-\alpha)}{a}} \mu(d\vec{\xi}) dt \\ &= 2^{n(2(b+r)-b\alpha/a)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt\end{aligned}$$

and now we apply Lemma 3.5 to get

$$\begin{aligned}\frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt &\leq \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} \left( \int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right)^2 d\vec{x} \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\vec{\xi})|^2 d\vec{\xi} \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} T_n\end{aligned}$$

and this proves the first inequality. An application of Lemma 3.6 gives is the second inequality.

**Nick's Question 2.8 Theorem 3.1** 1.6 has a unique solution  $\{u(t, x) : t > 0, x \in \mathbb{R}^d\}$  and

$$\sup_{(t, x) \in [0, T] \times \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) < \infty \quad \forall p \geq 2 \text{ and } T > 0.$$

**Proof:** First using 2.5 (details in notebook) we can show that

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\tilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n). \quad (2.9)$$

In order to prove the theorem we need to show that  $\sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$ . From the scaling property 2.6 we see that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \int_0^\infty e^{-t} t^{[2(b+r) - \frac{b\alpha}{a}]n} dt \\ &= \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \end{aligned}$$

as long as we have

$$2(b+r) - \frac{b\alpha}{a} > 0 \quad (2.10) \quad \text{alpha3}$$

By using Lemma 3.7 we can deduce that

$$\begin{aligned} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt}{\Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\ &\leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{\Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\ &\leq \frac{\left( 2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{\Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \end{aligned}$$

By the same reasoning as in the proof in the wave paper (details in my notebook) we can say that there exists a constant depending on  $\alpha$ ,  $k_\alpha$ , such that

$$\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \leq k_\alpha \frac{\left( 2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - b\alpha/a}}$$

Therefore,

$$\begin{aligned} \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 &\leq \sum_{n \geq 0} \theta^n n! k_\alpha \frac{\left( 2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - b\alpha/a}} \\ &= \sum_{n \geq 0} \theta^n k_\alpha \frac{\left( 2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - 1 - b\alpha/a}} \end{aligned}$$

and the last sum is finite when

$$2(b+r) - 1 - b\alpha/a > 0 \quad (2.11) \quad \text{alpha4}$$

Some definitions and results for the following lemma which come from the Xia Chen paper.

$$\rho = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 dx$$

$$\mathcal{E}_a(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |x|^a |\mathcal{F}(x)|^2 dx$$

$$\begin{aligned} \Lambda_\alpha &= \sup_{f \in \mathcal{F}_a} \left\{ \left( \int_{\mathbb{R}^{2d}} \frac{f^2(x)f^2(y)}{|x+y|^\alpha} dx dy \right)^{1/2} - \mathcal{E}_a(f, f) \right\} \\ &= \sup_{f \in \mathcal{F}_a} \left\{ \langle f^2 * f^2, |\cdot|^{-\alpha} \rangle^{1/2} - \mathcal{E}_a(f, f) \right\} \end{aligned}$$

The following is the correct representation for  $\rho$  from Theorem 1.5 in the Xia Chen paper. It fixes the  $(2\pi)^{-d}$  typo which I have the work shown above. See the work leading up to [1.14](#).

$$\rho = \Lambda_\alpha^{(2a-\alpha)/a}$$

Linking the above to the Le Chen notation from wave paper, we define

$$\mathcal{M}_a(\Theta f, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, \Theta f \rangle^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\}$$

and we can show that

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(f, 1)$$

and so

$$\Lambda_\alpha = \mathcal{M}_a(|\cdot|^{-a}, 2) = 2^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(|\cdot|^{-a}, 1)$$

which leads us to the following variational representation for  $\rho$

$$\rho = \left[ 2^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(|\cdot|^{-a}, 1) \right]^{(2a-\alpha)/a} = 2^{-\alpha/a} \mathcal{M}_a(|\cdot|^{-a}, 1)^{(2a-\alpha)/a} := 2^{-\alpha/a} \mathcal{M}_a^{(2a-\alpha)/a} \quad (2.12) \quad \boxed{\text{rhorep}}$$

**Definition of  $\rho_{new}$  which can be applied to the case of general  $\nu > 0$  and not just  $\nu = 2$**

Restating what was proven in [1.15](#), for the case of general  $\nu$ , we define

$$\rho_{new} := \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx)$$

It is proven in [1.15](#) that

1.  $\rho_{new} = \left(\frac{\nu}{2}\right)^{-\alpha/a} \rho$
2.  $\rho_{new} = \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}$
3.  $\lim_{n \rightarrow \infty} \left( \frac{1}{(n!)^2} T_n \right) \rightarrow \log(\rho_{new})$
- 4.

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n) = \log \rho_{new}(\mathcal{F}f)$$

where

$$\rho_{new}(g) = \sup_{\|h\|_{L_2}=1} \int_{\mathbb{R}^d} g(\xi) \left[ \int_{\mathbb{R}^d} \frac{h(\xi+\eta)h(\eta)}{\sqrt{1+\frac{\nu}{2}|\xi+\eta|^a} \sqrt{1+\frac{\nu}{2}|\eta|^a}} d\eta \right] \mu(d\xi)$$

**Nick's Question 2.9 Lemma 4.3**

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) = \log \left( 2^{2(b+r) - \frac{b\alpha}{a}} \right) + \log \left[ \left( \frac{\nu}{2} \right)^{-\alpha/a} \rho \right] \quad (2.13) \quad \boxed{2ineqa}$$

$$= \log \left( 2^{2(b+r) - \frac{b\alpha}{a}} \right) + \log [\rho_{new}] \quad (2.14)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( (n!)^{2(b+r) - \frac{b\alpha}{a}} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[ \left( \frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \right] + \log \left[ \left( \frac{\nu}{2} \right)^{-\alpha/a} \rho \right] \quad (2.15) \quad \boxed{2ineqb}$$

$$= \log \left[ \left( \frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \right] + \log [\rho_{new}] \quad (2.16)$$

Also, see the end of the proof for an alternate representation of 2.13 with  $\rho$  replaced with 2.12. **proof:** Denote  $T'_n$  to be  $T_n$  with  $\nu = 2$ , in other words

$$T'_n = \int_{(\mathbb{R}^d)^n} \left[ \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_j \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n).$$

Recall Theorem 2.3 from the Xia Chen paper which says that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{1}{(n!)^2} T'_n \right) \rightarrow \log(\rho)$$

Note that through a change of variables,  $\xi_i = \left(\frac{\nu}{2}\right)^{1/a} \xi_i$ , we can rewrite  $T_n = \left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n$ . We already showed in the proof of Theorem 3.1 above that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)$$

and then Lemma 3.7 above shows that

$$\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \leq \frac{2^{n \left[ 2(b+r) - \frac{b\alpha}{a} \right]}}{(n!)^2} T_n = \frac{2^{n \left[ 2(b+r) - \frac{b\alpha}{a} \right]}}{(n!)^2} \left( \frac{\nu}{2} \right)^{-n\alpha/a} T'_n.$$

Combining this and Theorem 2.3 from the Xia Chen paper we see that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \left( \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) \\ \leq \log \left( \left( \frac{\nu}{2} \right)^{-\alpha/a} 2^{\left[ 2(b+r) - \frac{b\alpha}{a} \right]} \right) + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{T'_n}{(n!)^2} \right) \\ = \log \left( \left( \frac{\nu}{2} \right)^{-\alpha/a} 2^{\left[ 2(b+r) - \frac{b\alpha}{a} \right]} \right) + \log(\rho) \end{aligned}$$

which proves the upper bound for [2.13](#).

Now define for  $t > 0$  and  $t' > 0$

$$J_n(t, t') = \int_{\mathbb{R}^{nd}} H_n(t, x) H_n(t', x) dx$$

and recall Lemma 3.4 which says

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{\mathbb{R}^{nd}} H_n(t, x)^2 dx = \frac{1}{(n!)^2} J_n(t, t).$$

Thus  $J_n(t, t)$  satisfies the scaling property

$$J_n(t, t) = t^{[2(b+r)-b\alpha/a]n} J_n(1, 1).$$

Let  $\tau$  and  $\tilde{\tau}$  be independent exponential random variables with mean 1. By lemma 3.3,

$$\begin{aligned} \mathbb{E}[J_n(\tau, \tau')] &= \int_0^\infty \int_0^\infty e^{-t} e^{-\tilde{t}} J_n(t, \tilde{t}) dt d\tilde{t} = \int_{\mathbb{R}^{dn}} \left[ \int_0^\infty e^{-t} H_n(t, x) dt \right]^2 dx \\ &= \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \cdots \mu(d\xi_n) = T_n \end{aligned}$$

In addition,

$$J_n(t, t') \leq J_n(t, t)^{1/2} J_n(t', t')^{1/2} = t^{[2(b+r)-b\alpha/a]n/2} t'^{[2(b+r)-b\alpha/a]n/2} J_n(1, 1).$$

Combining all of this gives us

$$\begin{aligned} T_n &= \mathbb{E}[J_n(\tau, \tilde{\tau})] \leq \mathbb{E} \left[ \tau^{[2(b+r)-b\alpha/a]n/2} \right] \mathbb{E} \left[ \tilde{\tau}^{[2(b+r)-b\alpha/a]n/2} \right] J_n(1, 1) \\ &= \Gamma \left( \frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 J_n(1, 1) \\ &= \Gamma \left( \frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 (n!)^2 \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \end{aligned}$$

where the last line follows from Lemma 3.4. Thus so far we have shown

$$\frac{T_n}{(n!)^2} = \frac{\left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n}{(n!)^2} \leq \Gamma \left( \frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad (2.17) \quad \boxed{2lowbd}$$

Now, another application of Stirling's formula shows that

$$\Gamma \left( \frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \sim \Gamma([2(b+r)-b\alpha/a]n+1) 2^{-[2(b+r)-b\alpha/a]n} C_n$$

where  $C_n = 2^{-1}[2(b+r)-b\alpha/a]^{1/2}(2\pi n)^{1/2}$ . We now infer from [2.17](#) that

$$c_\alpha \frac{2^{[2(b+r)-b\alpha/a]n}}{C_n} \frac{1}{(n!)^2} \left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n \leq \Gamma([2(b+r)-b\alpha/a]n+1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

where  $c_\alpha$  is a constant depending on  $\alpha$ . Since  $\frac{1}{c}C_n \rightarrow 0$  as  $n \rightarrow \infty$ , we obtain the lower bound

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{1}{n} \log \left( \Gamma([2(b+r) - b\alpha/a]n + 1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) \\ \geq \log \left( \left( \frac{\nu}{2} \right)^{-\alpha/a} 2^{[2(b+r) - b\alpha/a]} \right) + \log(\rho) \end{aligned}$$

Thus 2.13 is proven. 2.15 is shown in the same way as in the wave paper. If we replace  $\rho$  with 2.12 then we get that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left( \left[ 2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) = \log \left( (2\pi)^{-d} (\nu)^{-\alpha/a} 2^{2(b+r) - \frac{b\alpha}{a}} \mathcal{M}_a^{(2a-\alpha)/a} \right) \quad (2.18) \quad \boxed{43lim}$$

Note that there is a typo in the rough draft where there the  $(2\pi)^{-d}$  term is missing.

### Nick's Question 2.10 Theorem 4.1 (p=2 case)

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ = \theta^{\frac{a}{2a(b+r) - b\alpha - a}} \left( \frac{\nu}{2} \right)^{\frac{-\alpha}{2a(b+r) - b\alpha - a}} \left( \frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r) - b\alpha - a}} \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \end{aligned} \quad (2.19)$$

$$= \theta^{\frac{a}{2a(b+r) - b\alpha - a}} \left( \frac{2a}{2a(b+r) - b\alpha} \right)^\beta \left( \left( \frac{\nu}{2} \right)^{-\alpha/a} \rho \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \quad (2.20)$$

$$= \theta^{\frac{a}{2a(b+r) - b\alpha - a}} \left( \frac{2a}{2a(b+r) - b\alpha} \right)^\beta (\rho_{new})^{\frac{a}{2a(b+r) - b\alpha - a}} \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \quad (2.21)$$

**Proof:**

$$\begin{aligned} \mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n (n!) \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n (n!) t^{(2(b+r) - b\alpha/a)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} x_n R_n t^{(2(b+r) - b\alpha/a)n} \end{aligned}$$

where

$$R_n = (n!)^{2(b+r) - b\alpha/a} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad x_n = \frac{\theta^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}}$$

2.15 above says that

$$\frac{1}{n} \log(R_n) \rightarrow \log \left[ \left( \frac{\nu}{2} \right)^{-\alpha/a} \left( \frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right] \quad \text{as } n \rightarrow \infty.$$

Now define  $R$  to be

$$R = \left( \frac{\nu}{2} \right)^{-\alpha/a} \left( \frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho.$$

We want to find a  $\beta$  and  $A$  so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} x_n R^n \left( t^{2(b+r)-b\alpha/a} \right)^n = A$$

By lemma A3 notice that

$$\lim_{t \rightarrow \infty} \left[ \frac{1}{(\theta R) t^{2(b+r)-b\alpha/a}} \right]^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r)-b\alpha/a}]^n}{(n!)^{2(b+r)-(b\alpha/a)-1}} = 2(b+r) - (b\alpha/a) - 1$$

and by an easy algebraic manipulation

$$\begin{aligned} \lim_{t \rightarrow \infty} \left[ \frac{1}{t^{2(b+r)-b\alpha/a}} \right]^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r)-b\alpha/a}]^n}{(n!)^{2(b+r)-(b\alpha/a)-1}} \\ = [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \end{aligned}$$

hence we see that

$$\beta = \frac{2(b+r) - b\alpha/a}{2(b+r) - (b\alpha/a) - 1} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{1}{2(b+r)-(b\alpha/a)-1}}$$

or by removing the fractions in the denominator

$$\beta = \frac{2a(b+r) - b\alpha}{2a(b+r) - (b\alpha) - a} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{a}{2a(b+r)-b\alpha-a}}$$

Finally, lemma A2 says that

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ = \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left( \frac{\nu}{2} \right)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r)-b\alpha-a}} \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \end{aligned} \quad (2.22) \quad \boxed{\text{2pest}}$$

### Inconsistencies with section 7 of momentindep paper on githib:

In section 7, you have written that from the wave paper's proof of theorem 4.1 we replace  $\beta$ ,  $3 - \alpha$  and  $R$  with

$$\frac{2(b+r) - b\alpha/a}{2(b+r) - (b\alpha/a) - 1} \quad \text{and} \quad 2(b+r) - (b\alpha/a) - 1 \quad \text{and} \quad \nu^{-\alpha/a} \left( \frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \mathcal{M}_a^{(2a-\alpha)/a}$$

respectively. This is what I came up with too but in the above proof I left  $\rho$  in instead of replacing it with 2.12 like you did. However, I believe that (7.15) from section 7 in the momentindep paper is incorrect. In the wave paper notice how the numerator of the exponent of  $\mathcal{M}$  in (2.10) happens to be the numerator of  $\beta$  from the wave paper. This is just a coincidence and does not happen in the fraction equation which we can see from section 7. Because of this we cannot do the same simplifying and include the  $\mathcal{M}$  term in the  $(\cdot)^\beta$  term as is shown in (7.15). Below I will simplify 2.22 by replacing  $\rho$  with 2.12.

$$2.22 = \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left( \frac{\nu}{2} \right)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a}{2a(b+r) - b\alpha} \right)^\beta \left( (2\pi)^{-d} 2^{-\alpha/a} \mathcal{M}_a^{(2a-\alpha)/a} \right)^{\frac{a}{2a(b+r)-b\alpha-a}}$$

$$\begin{aligned}
& \times \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \\
& = \theta^{\frac{a}{2a(b+r)-b\alpha-a}} (\nu)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a}{2a(b+r)-b\alpha} \right)^\beta \left( (2\pi)^{-d} \right)^{\frac{a}{2a(b+r)-b\alpha-a}} \left( \mathcal{M}_a^{(2a-\alpha)/a} \right)^{\frac{a}{2a(b+r)-b\alpha-a}} \\
& \quad \times \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \\
& = \left( (2\pi)^{-d} \theta \right)^{\frac{a}{2a(b+r)-b\alpha-a}} (\nu)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a}{2a(b+r)-b\alpha} \right)^\beta \mathcal{M}_a^{\frac{2a-\alpha}{2a(b+r)-b\alpha-a}} \\
& \quad \times \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \\
& = \left( (2\pi)^{-d} \theta \right)^{\frac{a}{2a(b+r)-b\alpha-a}} (\nu)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a \mathcal{M}_a^{\frac{2a-\alpha}{2a(b+r)-b\alpha-a}}}{2a(b+r)-b\alpha} \right)^\beta \\
& \quad \times \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right)
\end{aligned}$$

and thus we see that

$$\begin{aligned}
& \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\
& = \left( (2\pi)^{-d} \theta \right)^{\frac{a}{2a(b+r)-b\alpha-a}} (\nu)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a \mathcal{M}_a^{\frac{2a-\alpha}{2a(b+r)-b\alpha-a}}}{2a(b+r)-b\alpha} \right)^\beta \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right)
\end{aligned} \tag{2.23}$$

2pest2

and this indeed simplifies to the wave case when  $a = b = 2$  and  $r = 0$ .

### Nick's Question 2.11 Theorem 5.1 (Upper Bound)

$$\begin{aligned}
& \limsup_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \\
& \leq \left( \frac{1}{2} \right) \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left( \frac{\nu}{2} \right)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a}{2a(b+r)-b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r)-b\alpha-a}} \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \\
& = \left( \frac{1}{2} \right) \left( (2\pi)^{-d} \theta \right)^{\frac{a}{2a(b+r)-b\alpha-a}} (\nu)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a \mathcal{M}_a^{\frac{2a-\alpha}{2a(b+r)-b\alpha-a}}}{2a(b+r)-b\alpha} \right)^\beta \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right)
\end{aligned}$$

where  $\beta$  is defined in Theorem 4.1.

**In terms of  $\rho_{new}$  we have**

$$\begin{aligned}
& \limsup_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \\
& \leq \left( \frac{1}{2} \right) \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left( \frac{\nu}{2} \right)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left( \frac{2a}{2a(b+r)-b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r)-b\alpha-a}} \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \\
& = \left( \frac{1}{2} \right) \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left( \frac{2a}{2a(b+r)-b\alpha} \right)^\beta \left( \left( \frac{\nu}{2} \right)^{-\alpha/a} \rho \right)^{\frac{a}{2a(b+r)-b\alpha-a}} \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right)
\end{aligned}$$

$$= \left(\frac{1}{2}\right) \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left(\frac{2a}{2a(b+r)-b\alpha}\right)^\beta (\rho_{new})^{\frac{a}{2a(b+r)-b\alpha-a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right)$$

**proof:** We denote  $u_\lambda$  for  $\lambda > 0$  to be the the solution of the fractional equation with  $\theta$  replaced with  $\lambda$  and  $u = u_\theta$ . Using theorem B1 from the wave paper we can say that for any  $p \geq 2$  that

$$\|u(t, 0)\|_p \leq \|u_{(p-1)\theta}(t, 0)\|_2$$

and by the scaling property

$$\begin{aligned} \|u_{(p-1)\theta}(t, 0)\|_2^2 &= \sum_{n \geq 0} (n!)^n \theta^n (p-1)^n \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (n!)^n \theta^n \left\| \tilde{f}_n\left(\cdot, 0, t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}}\right) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \left\| u\left(t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}}, 0\right) \right\|_2^2 \end{aligned}$$

and from this we can see that

$$\|u(t, 0)\|_p \leq \left\| u\left(t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}}, 0\right) \right\|_2 =: \|u(t_p, 0)\|_2$$

and an application of Theorem 4.1 proves the result.

Function needed for lower bound

$$W_n(t, \phi) := \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}Y(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds$$

**Nick's Question 2.12 Lemma 6.2**

$$\int_0^\infty e^{-t} W_n(t, \phi) dt = \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi) \dots \mu(d\xi_n)$$

**proof:** Same proof as in wave paper. details in notebook

**Nick's Question 2.13 Proposition 6.3**

For  $f \in \mathcal{H}$ ,  $t > 0$ , and  $p, q > 1$  with  $p^{-1} + q^{-1} = 1$ , it holds that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2}(q-1) \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|$$

**proof:** Same as wave paper.

In order to give a fractional version of Lemma 6.4 and Proposition 6.5, we need to derive the fractional version of the time scale distribution given at the bottom of page 25 and top of page 26 of the wave paper (equation 2.6). We start with a the general form

$$f_t(x) = t^V f(t^W x)$$

Note that for  $\xi \in \mathbb{C}^d$

$$\mathcal{F}f_t(\xi) = t^{V-dW} \mathcal{F}f(t^{-W}\xi)$$

We follow out the steps of Lemma 6.4 using the  $f_t$  defined above and then end up at a series of two linear equations with  $V$  and  $W$  being the unknowns which allows us to find  $W$  and  $V$ . First we calculate  $\|f_\tau\|_{\mathcal{H}}^2$ .

$$\begin{aligned}\|f_\tau\|_H^2 &= \tau^{2(V-dW)} \int_{\mathbb{R}^d} |\mathcal{F}f(\tau^{-W}\xi)|^2 \varphi(\xi) d\xi \\ &= \tau^{2V-dW} \int_{\mathbb{R}^d} |\mathcal{F}f(u)|^2 \tau^{-W(d-\alpha)} \varphi(u) du \\ &= \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2\end{aligned}$$

Lastly, we calculate  $W_n(t, \mathcal{F}f_\tau)$

$$\begin{aligned}W_n(t, \mathcal{F}f_\tau) &= \tau^{n(V-dW)} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(\tau^{-W}\xi_k) \prod_{k=1}^n \mathcal{F}Y(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds \\ &= \tau^{n(V-W(d-\alpha))} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(u_k) \prod_{k=1}^n \mathcal{F}Y(s_k - s_{k-1}, \cdot)(\tau^W(u_k + \dots + u_n)) \varphi(\mathbf{u}) d\mathbf{u} ds \\ &= \tau^{n(V-W(d-\alpha))} \tau^{-n\frac{a}{b}W(b+r-1)} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(u_k) \prod_{k=1}^n \mathcal{F}Y(\tau^{\frac{aW}{b}}(s_k - s_{k-1}), \cdot)(u_k + \dots + u_n) \varphi(\mathbf{u}) d\mathbf{u} ds \\ &= \tau^{n(V-W(d-\alpha))} \tau^{-n\frac{a}{b}W(b+r-1)} \tau^{-n\frac{aW}{b}} W_n(t\tau^{\frac{a}{b}W}, \mathcal{F}f) \\ &= \tau^{n[V-W((d-\alpha)+\frac{a}{b}(b+r))]} W_n(t\tau^{\frac{a}{b}W}, \mathcal{F}f)\end{aligned}$$

In order to ensure Lemma 6.4 and Proposition 6.5 works in the fractional case we need to choose  $V$  and  $W$  so that

$$\begin{cases} V - W((d-\alpha) + \frac{a}{b}(b+r)) = 0 \\ 2(V - dW) + W\alpha = \beta \end{cases}$$

Doing this ensures that

$$\|f_t\|_{\mathcal{H}}^2 = t^\beta \|f\|_{\mathcal{H}}^2$$

and

$$W_n(t, \mathcal{F}f_\tau) = W_n(t\tau^{\beta-1}, \mathcal{F}f)$$

If we solve the above system of equations for  $V$  and  $W$  we see that

$$\begin{cases} V = (\frac{a}{b}(b+r) - \alpha + d) \frac{b}{a}(\beta-1) \\ W = \frac{b}{a}(\beta-1) \end{cases}$$

and with all of this we have the following fractional version of Lemma 6.4.

**Nick's Question 2.14 Lemma 6.4** For

$$f_t(x) = t^{(\frac{a}{b}(b+r) - \alpha + d) \frac{b}{a}(\beta-1)} f\left(t^{\frac{b}{a}(\beta-1)} x\right)$$

we have that

$$\|f_\tau\|_{\mathcal{H}}^2 = \tau^\beta \|f\|_{\mathcal{H}}^2$$

and also

$$W_n(t, \mathcal{F}f_\tau) = W_n(t\tau^{\beta-1}, \mathcal{F}f)$$

Note that the above Lemma does reduce down to the wave equation. IE when  $a = b = 2$  and  $r = 0$  and  $\beta = (4 - \alpha)/(3 - \alpha)$  then the above reduces down to  $f_t$  defined on the top of page 26 in the wave paper.

**Nick's Question 2.15 Proposition 6.5** that

For  $p, q > 1$  and  $p^{-1} + q^{-1} = 1$  we have

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|$$

**proof:** We apply proposition 6.3 to  $f_\tau$  with  $\tau = (p-1)t$  and use Lemma 6.4 which gives us

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} (q-1)(p-1)^\beta t^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t([p-1]t)^{\beta-1}, \mathcal{F}f) \right|.$$

which finishes the proof since

$$(q-1)(p-1)^\beta t^\beta = (p-1)^{\beta-1} t^\beta = t_p^\beta$$

Some preparation for Theorem 6.6. Recall that

$$W_n(t, \phi) := \int_{[0, t]^{n <}} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}Y(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds$$

and we now consider the space  $\mathcal{H}_+ := \mathcal{H} \cap C_+(\mathbb{R}^d)$  where  $C_+$  is the space of continuous non-negative and nonnegative definite functions on  $\mathbb{R}^d$ . For any  $f \in \mathcal{H}_+$  we can show that

$$W_n(t, \mathcal{F}(f)) = \int_{[0, t]^{n <}} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n (f * \gamma)(x_k) \prod_{k=1}^n Y(s_k - s_{k-1}, x_k - x_{k-1}) dx ds =: U_n(t, f)$$

Under assumptions on the parameters, the greens function for the fractional equation is non-negative hence under these conditions we get that  $W_n(t, \mathcal{F}(f)) \geq 0$  for all the functions  $f \in \mathcal{H}_+$ .

Now we define

$$W_n(t) := \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}(f)) \quad \text{and} \quad U_n(t) := \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} U_n(t, f)$$

and with this we see that

$$W_n(t) \geq U_n(t) \geq 0$$

**Nick's Question 2.16 Rough sketch of Theorem 6.6**

Mimicking the steps in the

wave paper we start by letting  $\tau$  be an exponential random variable with mean 1. With this,

$$\mathbb{E}(U_n(\tau)) = \int_0^\infty e^{-t} U_n(t) dt \geq \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \int_0^\infty e^{-t} U_n(t, f) dt$$

For an  $f \in \mathcal{H}_+$  and  $\|f\|_{\mathcal{H}} = 1$  we have that  $\mathcal{F}(f)$  is even and in addition, Lemma 6.2 gives us that

$$\int_0^\infty e^{-t} U_n(t, f) dt = \int_0^\infty e^{-t} W_n(t, \mathcal{F}f) dt = \int_{\mathbb{R}^{dn}} \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n)$$

Using the info from 1.15, we can say that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n) = \log[\rho_{new}(\mathcal{F}f)]$$

where

$$\rho_{new}(g) = \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} g(\xi) \left[ \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta \right] \mu(d\xi)$$

and with this being said, we see that for any  $f \in \mathcal{H}_+$  and  $\|f\|_H = 1$

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(t)) \geq \log \rho(\mathcal{F}f)$$

We now need to calculate

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho_{new}(\mathcal{F}f)$$

Consider a function  $h \in L^2(\mathbb{R}^d)$ . The function  $h(\cdot)/\sqrt{1 + \frac{\nu}{2}|\cdot|^a} \in L^2(\mathbb{R}^d)$  so that  $g(x) := (2\pi)^{d/2} \mathcal{F}^{-1} \left( \frac{h(\cdot)}{\sqrt{1 + \frac{\nu}{2}|\cdot|^a}} \right) (x)$  is well defined. **I still need to verify the details that** under these conditions,  $g \in W_{1,a}(\mathbb{R}^d)$  with  $\|g\|_{W_{1,a}} = 1$ . Here is a sketch of the details

$$\begin{aligned} \|g\|_{W_{1,a}}^2 &= \|g\|_{L^2}^2 + (2\pi)^{-d} \sum_{k=1}^d \int_{\mathbb{R}^d} \xi_k^a |\mathcal{F}g(\xi)|^2 d\xi \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} (1 + |\xi|^a) |\mathcal{F}g(\xi)|^2 d\xi \\ &= \int_{\mathbb{R}^d} h(\xi)^2 d\xi = 1 \end{aligned}$$

where above we used the fact that  $\|g\|_{L^2}^2 = (2\pi)^{-d} \|\mathcal{F}g\|_{L^2}^2$ .

Now, for notation let  $\tilde{h}(x) = h(-x)$ . Note that

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}g(\xi - \eta) \mathcal{F}g(-\eta) d\eta \\ &= (2\pi)^{-d} (\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) \end{aligned}$$

Since  $h(\cdot)$  is real value then  $\widetilde{\mathcal{F}g} = \overline{\mathcal{F}g} = \mathcal{F}\bar{g}$ . Using this and the fact that  $\mathcal{F}(fg) = (2\pi)^{-d} \mathcal{F}(f) * \mathcal{F}(g)$ , we see that  $(2\pi)^{-d} (\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) = \mathcal{F}[|g|^2](\xi)$  and with this we see that

$$\begin{aligned} \rho_{new}(\mathcal{F}f) &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) (\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) \mathcal{F}[|g|^2](\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \langle f, |g|^2 \rangle_{\mathcal{H}}. \end{aligned}$$

With this we see that

$$\sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \rho_{new}(\mathcal{F}f) = \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \sup_{\|h\|_{L^2}=1} \langle f, |g|^2 \rangle_{\mathcal{H}} = \sup_{\|h\|_{L^2}=1} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2}$$

where the optimal  $f$  is chosen to be  $|g|^2 / \| |g|^2 \|_{\mathcal{H}}$ . **Can the optimal f be chosen like this? This is true for  $dx$  but is it true for  $\mu(dx)$ ? I believe that it would be true for  $\mu$  such that  $\mu$  has a density which is radially symmetric about the origin.**

**The Riesz kernal would then still be ok** Therefore the supremum can be restricted to  $\mathcal{H}_+$  which implies

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho_{new}(\mathcal{F}f) = \sup_{\|h\|_{L^2}=1} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2}$$

and notice that

$$\begin{aligned} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2} &= \left( \int_{\mathbb{R}^d} |\mathcal{F}[|g|^2](\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= (2\pi)^{-d} \left( \int_{\mathbb{R}^d} |(\mathcal{F}g * \tilde{\mathcal{F}}g)(\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= \left( \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta \right]^2 \mu(d\xi) \right)^{1/2} \\ &= \rho_{new}^{1/2} \end{aligned}$$

which finally leads us to the equality

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho_{new}^{1/2})$$

**Nick's Question 2.17 Lemma 6.7** We have that

$$W_n(t) = t^{n \frac{b}{a} (a - \frac{\alpha}{2} + \frac{a}{b} r)} W_n(1)$$

and moreover

$$U_n(t) = t^{n \frac{b}{a} (a - \frac{\alpha}{2} + \frac{a}{b} r)} U_n(1)$$

**proof:** Let  $\phi = \mathcal{F}f$  and  $c > 0$

$$\begin{aligned} W_n(t, \phi(c^{-1} \cdot)) &= \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(c^{-1} \xi_k) \prod_{k=1}^n Y(s_k - s_{k-1}, \cdot) (\xi_k + \cdots \xi_n) \varphi(\vec{\xi}) d\vec{\xi} d\vec{s} \\ &= c^{nd} \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n Y(s_k - s_{k-1}, \cdot) (c(\xi_k + \cdots \xi_n)) \varphi(c\vec{u}) d\vec{u} d\vec{s} \\ &= c^{nd} c^{-n(d-\alpha)} c^{-n \frac{a}{b} (b+r-1)} \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n Y\left(c^{\frac{b}{a}}(s_k - s_{k-1}), \cdot\right) (\xi_k + \cdots \xi_n) \varphi(c\vec{u}) d\vec{u} d\vec{s} \\ &= c^{nd} c^{-n(d-\alpha)} c^{-n \frac{a}{b} (b+r-1)} c^{-n \frac{a}{b}} \int_{[0, c^{\frac{a}{b}} t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n Y(s_k - s_{k-1}, \cdot) (\xi_k + \cdots \xi_n) \varphi(c\vec{u}) d\vec{u} d\vec{s} \\ &= c^{-n(a-\alpha+\frac{a}{b}r)} W_n\left(c^{\frac{a}{b}} t, \phi\right) \end{aligned}$$

Now we apply this with  $c^{\frac{a}{b}} = \frac{1}{t}$  to get

$$W_n\left(t, \phi\left(t^{\frac{b}{a}} \cdot\right)\right) = t^{n \frac{b}{a} (a - \alpha + \frac{a}{b} r)} W_n(1, \phi)$$

Now note that

$$\left\| \phi\left(t^{\frac{b}{a}} \cdot\right) \right\|_{L^2(\mu)}^2 = t^{-\frac{\alpha b}{a}} \|\phi\|_{L^2(\mu)}^2$$

and from this we deduce that

$$\left\| t^{\frac{\alpha b}{2a}} \phi(t^{\frac{b}{a}} \cdot) \right\|_{L^2(\mu)}^2 = \|\phi\|_{L^2(\mu)}^2 = \|f\|_{\mathcal{H}}^2$$

and in addition

$$W_n \left( t, t^{\frac{b\alpha}{2a}} \phi(t^{\frac{b}{a}} \cdot) \right) = t^{n\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} W_n(1, \phi)$$

and the result follows after taking the sup over all  $f \in \mathcal{H}$  with  $\|f\|_{\mathcal{H}} = 1$ .

**Nick's Question 2.18 Lemma 6.8**

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[ (n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \right] \geq \log \left[ \frac{\rho_{new}^{1/2}}{\left( \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right]$$

**proof:** Let  $\tau$  be an exponential random variable with mean 1. By lemma 6.7,

$$\mathbb{E}(U_n(\tau)) = \mathbb{E} \left( \tau^{n\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \right) = \Gamma \left( n\frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1)$$

and by Theorem 6.6 we obtain

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[ \Gamma \left( n\frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1) \right] \geq \log(\rho_{new}^{1/2}). \quad (2.24) \quad \boxed{1b1}$$

Recall that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{\Gamma(an+1)}{(n!)^a} \right) = a \log(a)$$

and with this we have that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[ \frac{\Gamma \left( n\frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right)}{(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right] = \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log \left( \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right)$$

and the result is proven after multiplying and dividing  $(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}$  on the inside of the log in 2.24

**Nick's Question 2.19 Lemma 6.9**

For any  $k, \theta > 0$ , there exists a constant  $c_1 = c_1(\alpha, \mathcal{M}, k, \theta) > 0$  such that, by setting  $n_t = [c_1 t]$ , it holds that

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log(k^{n_t} \theta^{n_t/2} U_{n_t}(t)) \geq \left( k\sqrt{\theta} \right)^{\frac{2a}{2ab-\alpha b+2ar}} \left( \rho_{new}^{1/2} \right)^{\frac{2a}{2ab-\alpha b+2ar}} \quad (2.25) \quad \boxed{69ineq}$$

**proof:** Let  $\epsilon > 0$ . Lemma 6.8 guarantees the existence of an  $N_\epsilon > 0$  so for all  $n > N_\epsilon$

$$(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \geq \exp(n(\log(R) - \epsilon)) = R^n e^{-n\epsilon}$$

where

$$R = \frac{\rho_{new}^{1/2}}{\left( \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}}.$$

Now fix a  $c > 0$  and let  $n_t := [ct]$ . Notice that  $n_t \geq N_\epsilon$  for any  $t > t_\epsilon := (N_\epsilon + 1)/c$ .

Let  $t > t_\epsilon$ .

$$k^{n_t} \theta^{n_t/2} U_{n_t}(t) = k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t} U_{n_t}(1) \geq \frac{k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} R^{n_t} e^{-n_t \epsilon}$$

Notice that  $[ct]/t \rightarrow c$  as  $t \rightarrow \infty$  which means that  $n_t/t \rightarrow c$  as  $t \rightarrow \infty$ . With this we can say

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \left( k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) = c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left( k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) =: I(n_t)$$

Now,

$$\begin{aligned} I(n_t) &\geq c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left( (kR\sqrt{\theta})^{n_t} \frac{t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) + c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left[ \left( \frac{t}{n_t} \right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t} \frac{n_t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right] - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left( \frac{n_t^{n_t}}{(n_t)!} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) - c\epsilon \end{aligned}$$

and letting  $\epsilon$  tend to 0 we see that

$$I(n_t) \geq c \left[ \log(k\sqrt{\theta}R) - \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + \frac{b}{a} \left( a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right] =: h(c).$$

We can easily find the maximal  $c$  by solving  $h'(c) = 0$  and doing so we get  $c = (k\sqrt{\theta}R)^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}}$ . After plugging in this value for  $c$  and replacing  $R$  we arrive at the following inequality

$$I(n_t) \geq (k\sqrt{\theta})^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}} (\rho_{new}^{1/2})^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}}$$

and simplifying exponents gives us [2.25](#).

**Nick's Question 2.20 Theorem 6.1** For  $\nu = 2$  it holds that

$$\begin{aligned} \liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p & \tag{2.26} \\ & \geq \left( \frac{1}{2} \right) \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left( \frac{2a}{2a(b+r)-b\alpha} \right)^\beta \rho_{new}^{\frac{a}{2a(b+r)-b\alpha-a}} \left( 2(b+r) - \frac{b\alpha}{a} - 1 \right) \end{aligned}$$

**proof:** By proposition 6.5, for and  $p, q > 0$  with  $p^{-1} + q^{-1} = 1$  we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|.$$

We now take the supremum over all  $f \in \mathcal{H}_+$  with  $\|f\|_{\mathcal{H}} = k > 0$ . Recall that  $W_n(t, \phi) = k^n W_n(t, \phi/k)$  for  $\phi \in L^2(\mu)$  and nonnegativity of  $W_n(t, \cdot)$  on  $\mathcal{H}_+$ . Let  $c > 0$ .

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+ \|f\|_{\mathcal{H}}=k} \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f)$$

$$\begin{aligned}
&= \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \sum_{n \geq 0} k^n \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\
&\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f)
\end{aligned}$$

where  $n_t = \lfloor ct_p^\beta \rfloor$ . Now by choosing  $c$  as in Lemma 6.9 we get that

$$\begin{aligned}
\liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &\geq \liminf_{n \rightarrow \infty} \left( \frac{-\frac{1}{2} t_p^\beta k^2}{t_p^\beta} + \frac{1}{t_p^\beta} \log \left[ \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \right] \right) \\
&\quad - \frac{1}{2} k^2 + (k\sqrt{\theta})^{\frac{2a}{2ab-\alpha b+2ar}} (\rho_{new}^{1/2})^{\frac{2a}{2ab-\alpha b+2ar}} =: h(k)
\end{aligned}$$

By maximizing  $h$  we see that  $h$  is maximized at the point  $k^* = \left( \frac{2ab-\alpha b+2ar}{2aB} \right)^{\frac{2ab-\alpha b+2ar}{2a-2[2ab-\alpha b+2ar]}}$  where  $B = (\sqrt{\theta} \rho_{new}^{1/2})^{\frac{2a}{2ab-\alpha b+2ar}}$ . Inserting  $k^*$  into  $h$  gives us that

$$h(k^*) = B^\beta \left( \frac{2a}{2ab-\alpha b+2ar} \right)^\beta \left( \frac{2ab-\alpha b+2ar-a}{2a} \right)$$

and plugging in the value for  $B$  yields [2.26](#).

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